# 3. Requirements

## 3.1 Functional Requirements

| Req# | Requirement | Comments | Priority |
|---|---|---|---|
| BR_01 | **Administrator** shall be able to create, update, deactivate user accounts, and assign roles and permissions to users because user access must be managed securely and different users require different levels of system access. | Business Process = User Management | 1 |
| BR_02 | **Administrator** shall be able to modify system configuration settings because the system must adapt to operational and organizational requirements. | Business Process = System Configuration | 2 |
| BR_03 | **Administrator** shall be able to monitor system activity logs because user actions and system events must be tracked for security and auditing. | Business Process = System Monitoring | 2 |
| BR_04 | **Administrator** shall be able to manage airport reference data (gates, stands, airport codes) because operational data must remain accurate and up to date. | Business Process = Data Maintenance | 2 |
| BR_05 | **Operations Staff** shall be able to assign gates to scheduled flights because aircraft must be directed to the correct airport gate. | Business Process = Gate Management | 1 |
| BR_06 | **Operations Staff** shall be able to create and update flight schedules because airport operations depend on accurate scheduling information. | Business Process = Flight Scheduling | 1 |
| BR_07 | **Operations Staff** shall be able to assign aircraft stands because aircraft parking and ground operations must be coordinated efficiently. | Business Process = Stand Allocation | 1 |
| BR_08 | **Operations Staff** shall be able to monitor flight status (arrivals, departures, delays) because operational planning requires up-to-date flight information. | Business Process = Flight Monitoring | 2 |
| BR_09 | **Operations Staff** shall be able to update operational records for flights, gates, and stands because airport operations must be accurately documented. | Business Process = Operations Management | 2 |
| BR_10 | **Operations Staff** shall be able to detect scheduling conflicts between flights, gates, and stands because operational overlaps must be avoided. | Business Process = Conflict Detection | 1 |
| BR_11 | **Security Staff** shall be able to monitor airport surveillance cameras because continuous visual monitoring is required to detect suspicious activities and ensure passenger safety. | Business Process = Surveillance Monitoring | 1 |
| BR_12 | **Security Staff** shall be able to control and verify access to restricted airport areas because only authorized personnel | Business Process = Access Control | 1 |

| | | | |
|---|---|---|---|
| | should enter secure zones. | | |
| BR_13 | **Security Staff** shall be able to perform passenger and staff security checks because safety regulations require screening before entering secure airport areas. | Business Process = Security Screening | 1 |
| BR_14 | **Security Staff** shall be able to record and report security incidents because all security events must be documented for investigation and auditing. | Business Process = Incident Reporting | 2 |
| BR_15 | **Security Staff** shall be able to receive and respond to security alerts generated by the system because rapid response is required to prevent potential threats. | Business Process = Alert Management | 1 |
| BR_16 | **Security Staff** shall be able to review security logs and reports because monitoring historical security data helps identify risks and improve safety procedures. | Business Process = Security Monitoring | 2 |
| BR_17 | **Maintenance Staff** shall be able to record equipment maintenance activities because airport systems and equipment require regular servicing. | Business Process = Maintenance Logging | 1 |
| BR_18 | **Maintenance Staff** shall be able to schedule preventive maintenance tasks because regular inspections help avoid equipment failures. | Business Process = Preventive Maintenance | 1 |
| BR_19 | **Maintenance Staff** shall be able to report infrastructure issues such as damaged facilities or equipment because operational safety depends on well-maintained infrastructure. | Business Process = Issue Reporting | 1 |
| BR_20 | **Maintenance Staff** shall be able to track the status of maintenance requests because maintenance operations must be coordinated efficiently. | Business Process = Maintenance Tracking | 2 |
| BR_21 | **Maintenance Staff** shall be able to identify, assess, track, and document maintenance issues and repairs because a complete maintenance lifecycle requires continuous monitoring and accurate reporting from detection to resolution. | Business Process = Issue Management | 1 |
| BR-22 | **Ground handling staff shall view and manage the arrival schedule of aircraft, including assigned gates, parking stands, and estimated arrival times.** | Business Process = Schedule Monitoring | 2 |
| BR-23 | **Ground handling staff shall track, update, and manage baggage loading and unloading operations for each flight.** | Business Process = Baggage Handling | 1 |
| BR-24 | **Ground handling staff shall assign and monitor apron service tasks such as refueling coordination, catering delivery, and ground equipment allocation.** | Business Process = Apron Management | 1 |
| BR-25 | **Ground handling staff shall record and monitor aircraft turnaround activities to ensure readiness for departure** | Business Process = Departure Preparation | 2 |

| | | | |
|---|---|---|---|
| BR-26 | **Ground handling staff shall share flight status updates and operational information with airline staff and airport authorities.** | Business Process = Update Reporting | 2 |
| BR-27 | **Ground handling staff shall confirm completion of ground services (baggage loading, catering, fueling readiness) before an aircraft departure.** | Business Process = Confirmation Of Duty | 1 |
| BR-28 | **Airline staff shall perform passenger check-in, verify identification, and confirm booking details.** | Business Process = Passenger Verification | 1 |
| BR-29 | **Airline staff shall generate and issue boarding passes to checked-in passengers.** | Business Process = Pass Printing/Issuing | 2 |
| BR-30 | **Airline staff shall assign, modify, or reallocate passenger seats for a flight.** | Business Process = Seat Handling | 3 |
| BR-31 | **Airline staff shall validate passenger tickets and confirm eligibility for boarding.** | Business Process = Validation | 1 |
| BR-32 | **Airline staff shall create, update, and view passenger manifests for their airline's flights.** | Business Process = Monitoring Manifests | 2 |
| BR-33 | **Airline staff shall manage the boarding process, including scanning boarding passes and marking passengers as boarded.** | Business Process = Boarding Handling | 1 |
| BR-34 | **Finance Staff shall be able to generate invoices for airport services (landing fees, parking fees, passenger services) because financial transactions must be properly recorded and billed.** | Business Process = Billing & Invoicing | 1 |
| BR-35 | **Finance Staff shall be able to generate financial reports (revenue, expenses, profit) because management requires financial insights.** | Business Process = Financial Reporting | 2 |
| BR-36 | **HR Staff shall be able to manage employee records including personal details, roles, and departments because staff information must be maintained centrally.** | Business Process = Employee Management | 1 |
| BR-37 | **HR Staff shall be able to manage employee schedules and shifts because airport operations require continuous staffing.** | Business Process = Shift Scheduling | 1 |
| BR-38 | **Asset Management Staff shall be able to register and manage airport assets (vehicles, equipment, IT systems) because all assets must be tracked centrally.** | Business Process = Asset Registration | 1 |

### 3.2 Non-Functional Requirements

### 3.2.1 Product Requirements

Requirements which specify that the delivered product must behave in a particular way e.g. execution speed, reliability, etc.

1. Flight Information
   The system must provide real-time flight information, including departure times, arrival times, delays, and gate numbers.
2. Passenger Check-in
   Passengers must be able to check in online and receive a digital boarding pass.
3. Notifications
   The system must send notifications for flight updates such as delays, cancellations, and gate changes.
4. Airport Navigation
   The system must provide an interactive map to help passengers find gates, restaurants, and services inside the airport.
5. Airport Services
   The system must allow passengers to access additional services such as parking reservations and lounge bookings.

#### 3.2.1.1 Usability Requirements

1. The system interface shall be simple and user-friendly for airport staff and administrators.
2. The system shall provide clear navigation menus for accessing all major functions.
3. The system shall display error messages that clearly explain the problem and how to fix it.
4. New users shall be able to learn the basic functions within 30 minutes of training.

#### 3.2.1.2 Performance Requirements

1. The system should respond quickly.
2. The system should support many users at the same time.
3. The system should update flight information in real time

#### 3.2.1.3 Availability

1. The system shall be available 24 hours per day, 7 days per week.
2. Scheduled maintenance shall not exceed 2 hours per month.

#### 3.2.1.4 Security

1. The system should require login.
2. The system should protect user data.
3. The system should use role based access.

#### 3.2.1.5 Reliability

1. The system should work without crashes.
2. The system should save data automatically.
3. The system should backup data automatically.

#### 3.2.1.6 Scalability

1. The system shall support future expansion to multiple airports if required.
2. The system shall allow increasing the number of users and flights without major system redesign.

### 3.2.2 Organizational Requirements

1. The system shall comply with the software development standards and documentation guidelines defined by the airport IT department.
2. The system shall be implemented using technologies approved by the airport organization's IT policies.
3. The system shall maintain system activity logs according to the airport authority's internal auditing procedures.
4. The system shall support regular data backup procedures defined by the airport IT department.
5. The system shall allow administrators to configure user roles and permissions according to airport organizational policies.
6. The system shall support staff training procedures by providing user manuals and system documentation.
   The system shall follow the organization's data management policies for storing and maintaining passenger and flight information.
7. The system shall allow integration with the airport's internal information systems used by staff and management.

### 3.2.3 External Requirements

1. The system shall comply with international aviation regulations and standards applicable to airport operations.
2. The system shall comply with data protection laws regarding the storage and processing of passenger personal information.
3. The system shall support interoperability with airline information systems for exchanging flight data.
4. The system shall support integration with external security and immigration systems when required.
5. The system shall ensure secure communication with external airlines and booking systems.

6. The system shall comply with payment processing regulations when handling passenger payments.
7. The system shall support standard data exchange formats used in aviation systems.
8. The system shall ensure that passenger data is shared with authorized external systems only when permitted by law.

#### 3.2.3.1 System Integration Requirements

The Airport Management System must be able to interact and exchange data with other external systems used within the airport ecosystem. These may include airline reservation systems, baggage handling systems, security monitoring systems, and the Airport Operational Database

#### 3.2.3.2 Legislative and Regulatory Requirements

The system must comply with international and national aviation regulations as well as data protection laws. Passenger information, flight data, and staff records must be processed and stored according to applicable privacy and security regulations.

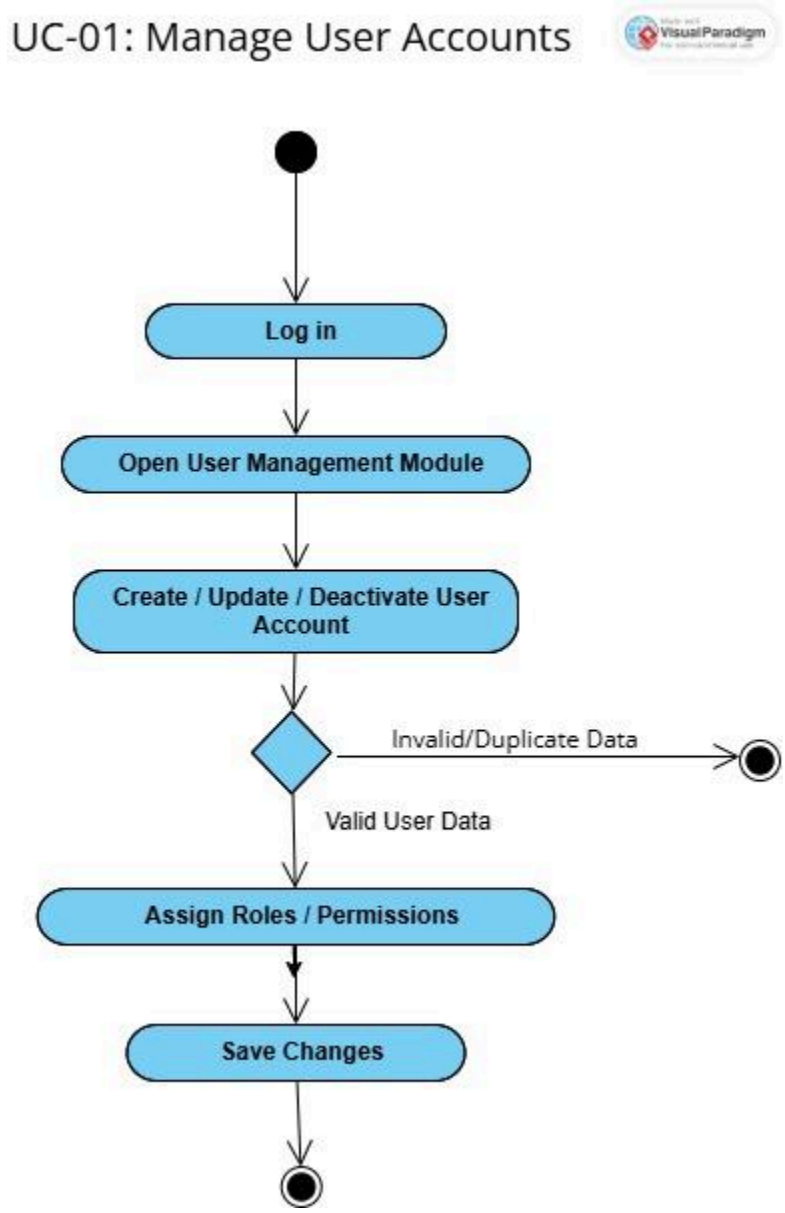
#### 3.2.3.3 Safety and Security Requirements

Due to the critical nature of airport operations, the system must implement strong security mechanisms. User authentication and authorization must be enforced based on defined user roles. Sensitive information such as passenger data, security logs, and operational details must be protected through secure communication protocols and data protection mechanisms.

**4. User Scenarios/Use Cases**

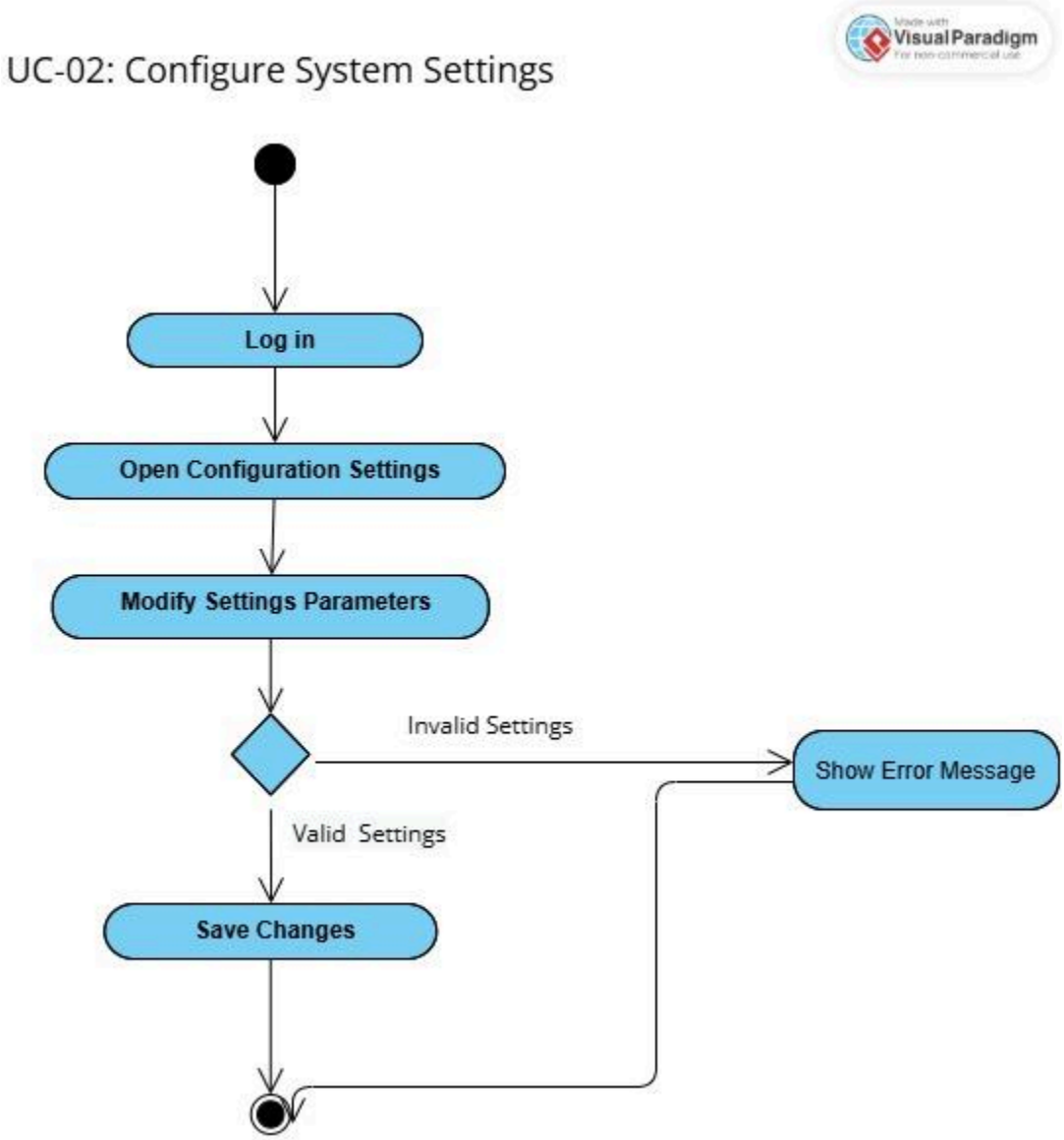
| UC Name | *UC-01: Manage User Accounts* |
|---|---|
| Summary | *Administrator manages user accounts, roles, and permissions.* |
| Dependency | *None* |
| Actors | ***Primary:*** *Administrator* |
| Preconditions | ● *Administrator is authenticated*<br>● *System is accessible* |
| Description of the Main Sequence | ● *Step 1: Administrator logs into the system*<br>● *Step 2: Navigates to user management module*<br>● *Step 3: Creates/updates/deactivates a user account*<br>● *Step 4: Assigns roles and permissions*<br>● *Step 5: Saves changes* |
| Description of the Alternative Sequence | ● *Step 1: Invalid user data entered → system shows error*<br>● *Step 2: Duplicate user detected → system prevents creation* |
| Non functional requirements | *Secure authentication*<br>*Role-based access control*<br>*Fast response time* |
| Postconditions | *User account is successfully/created/updated/deactivated* |

## Activity Diagram UC-01



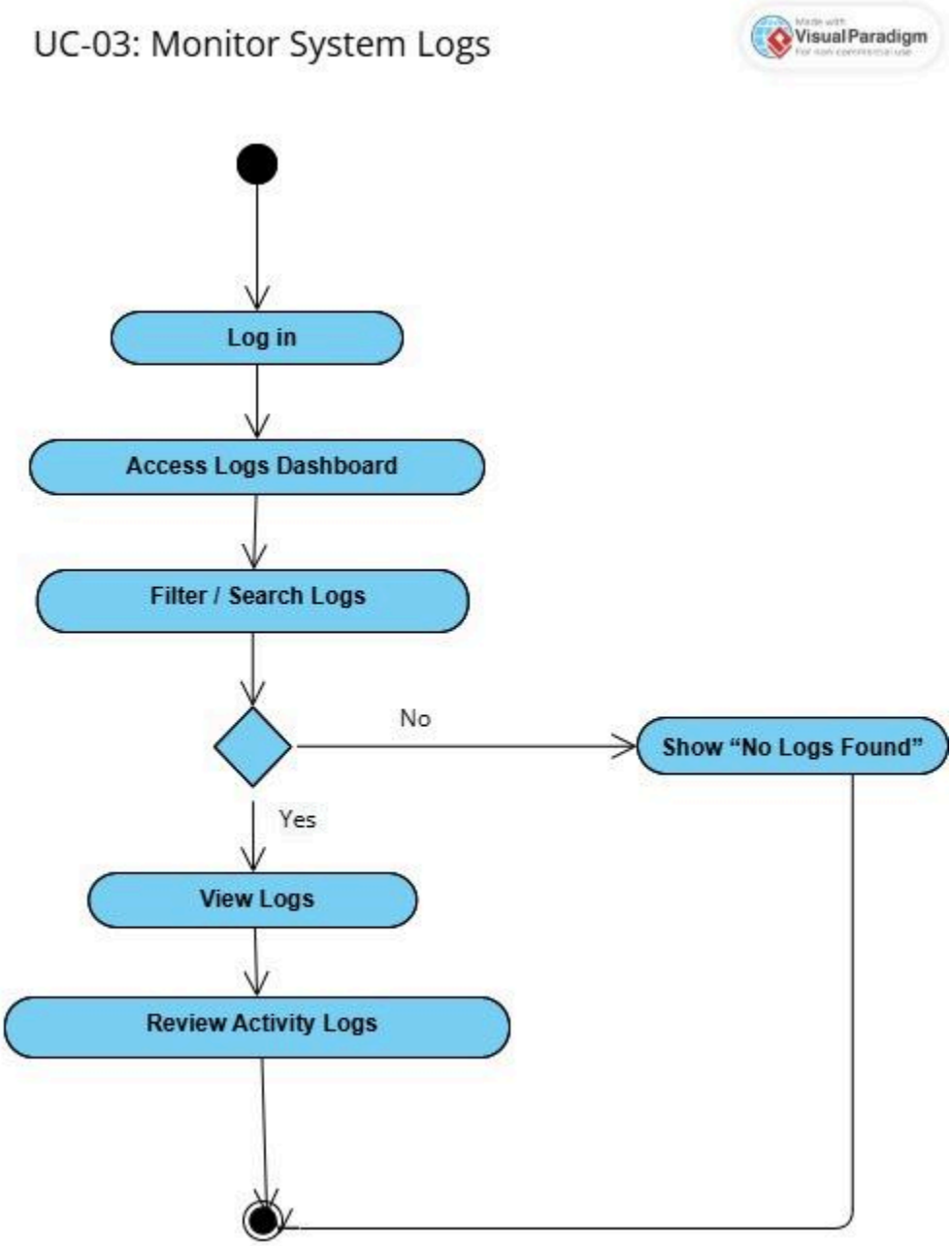
| UC Name | *UC-02: Configure System Settings* |
|---|---|
| Summary | *Administrator modifies system configuration settings.* |
| Dependency | *UC-01* |
| Actors | ***Primary:*** *Administrator* |
| Preconditions | ● *Administrator is logged in* |
| Description of the Main Sequence | ● *Step 1: Open configuration settings*<br>● *Step 2: Modify required parameters*<br>● *Step 3: Save changes* |
| Description of the Alternative Sequence | ● *Step 1: Invalid configuration → system rejects input* |
| Non functional requirements | *System stability*<br>*Configuration validation* |
| Postconditions | *System settings updated successfully* |

## Activity Diagram UC-02



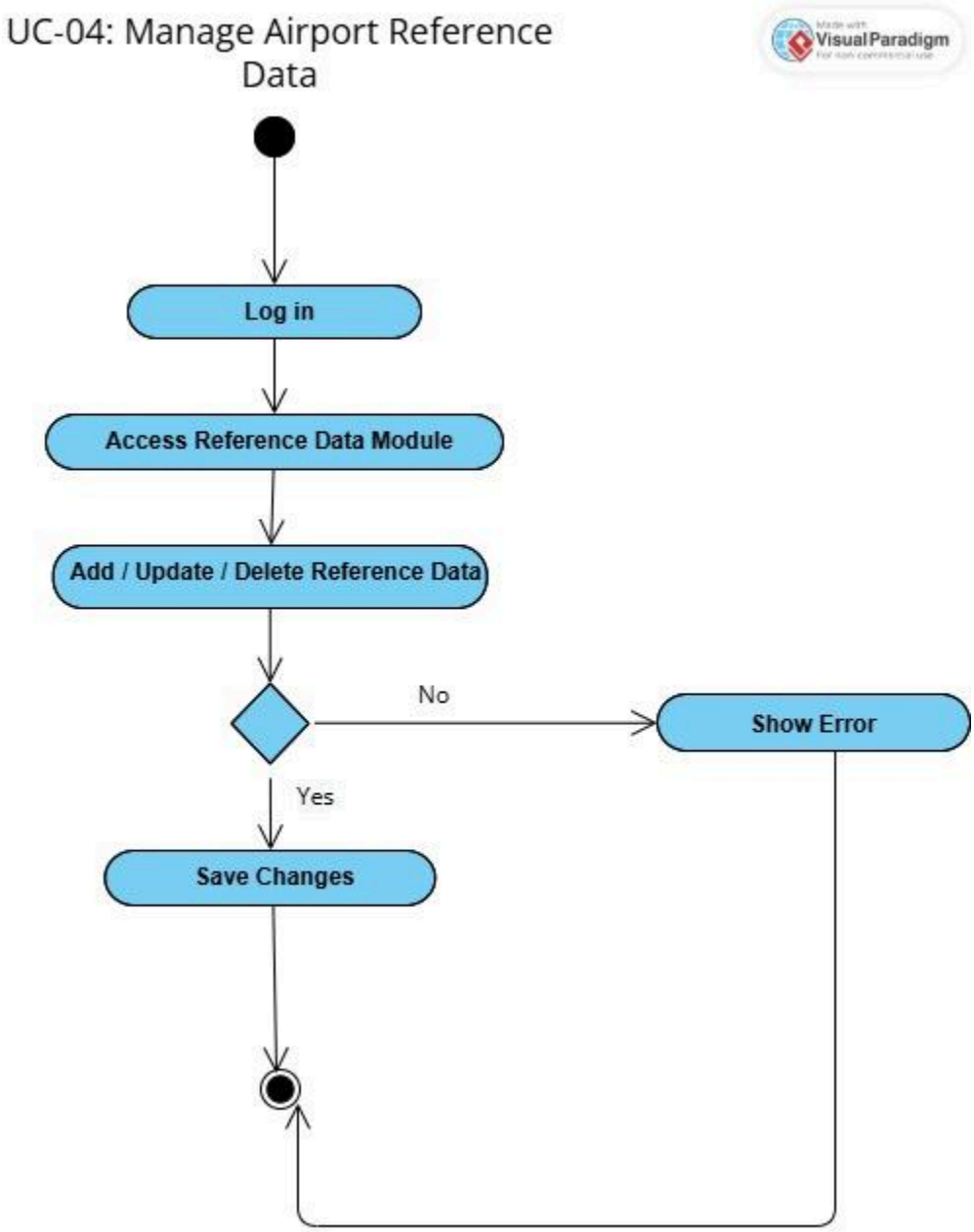
| UC Name | *UC-03: Monitor System Logs* |
|---|---|
| Summary | *Administrator views and monitors system activity logs* |
| Dependency | *None* |
| Actors | ***Primary:*** *Administrator* |
| Preconditions | ● *Admin access granted* |
| Description of the Main Sequence | ● *Step 1: Open logs dashboard*<br>● *Step 2: Filter/search logs*<br>● *Step 3: Review system activities* |
| Description of the Alternative Sequence | ● *Step 1: No logs found → system shows empty result* |
| Non functional requirements | *High performance log retrieval*<br>*Secure log storage* |
| Postconditions | *Logs reviewed* |

## Activity Diagram UC-03



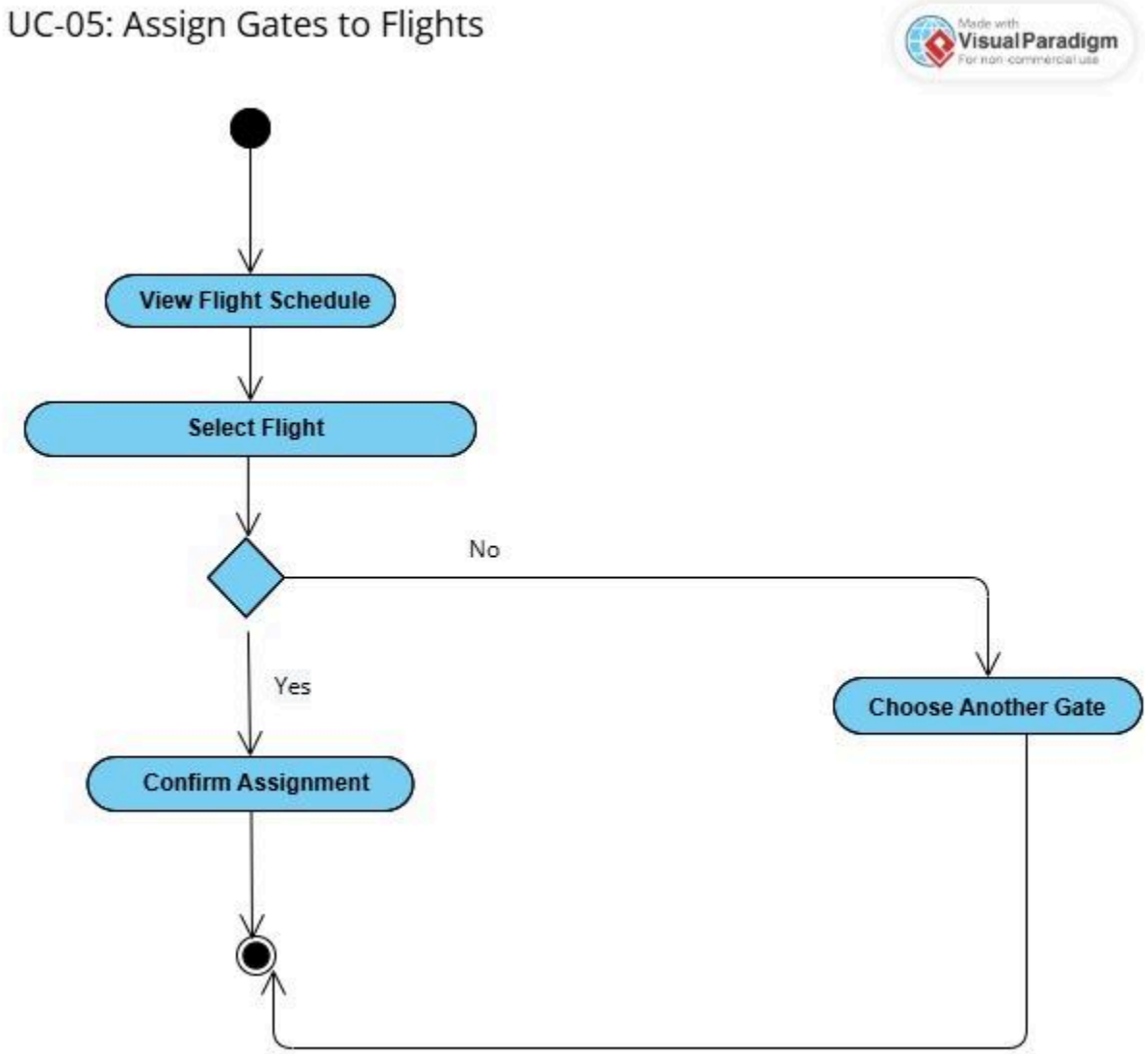
| UC Name | *UC-04: Manage Airport Reference Data* |
|---|---|
| Summary | *Administrator manages airport reference data (gates, stands, codes).* |
| Dependency | *UC-02* |
| Actors | ***Primary:*** *Administrator* |
| Preconditions | • *Admin logged in* |
| Description of the Main Sequence | • *Step 1: Access reference data module*<br>• *Step 2: Add/update/delete data*<br>• *Step 3: Save changes* |
| Description of the Alternative Sequence | • *Step 1: Invalid data → system shows error* |
| Non functional requirements | *Data integrity*<br>*Validation rules* |
| Postconditions | *Reference data updated* |

## Activity Diagram UC-04



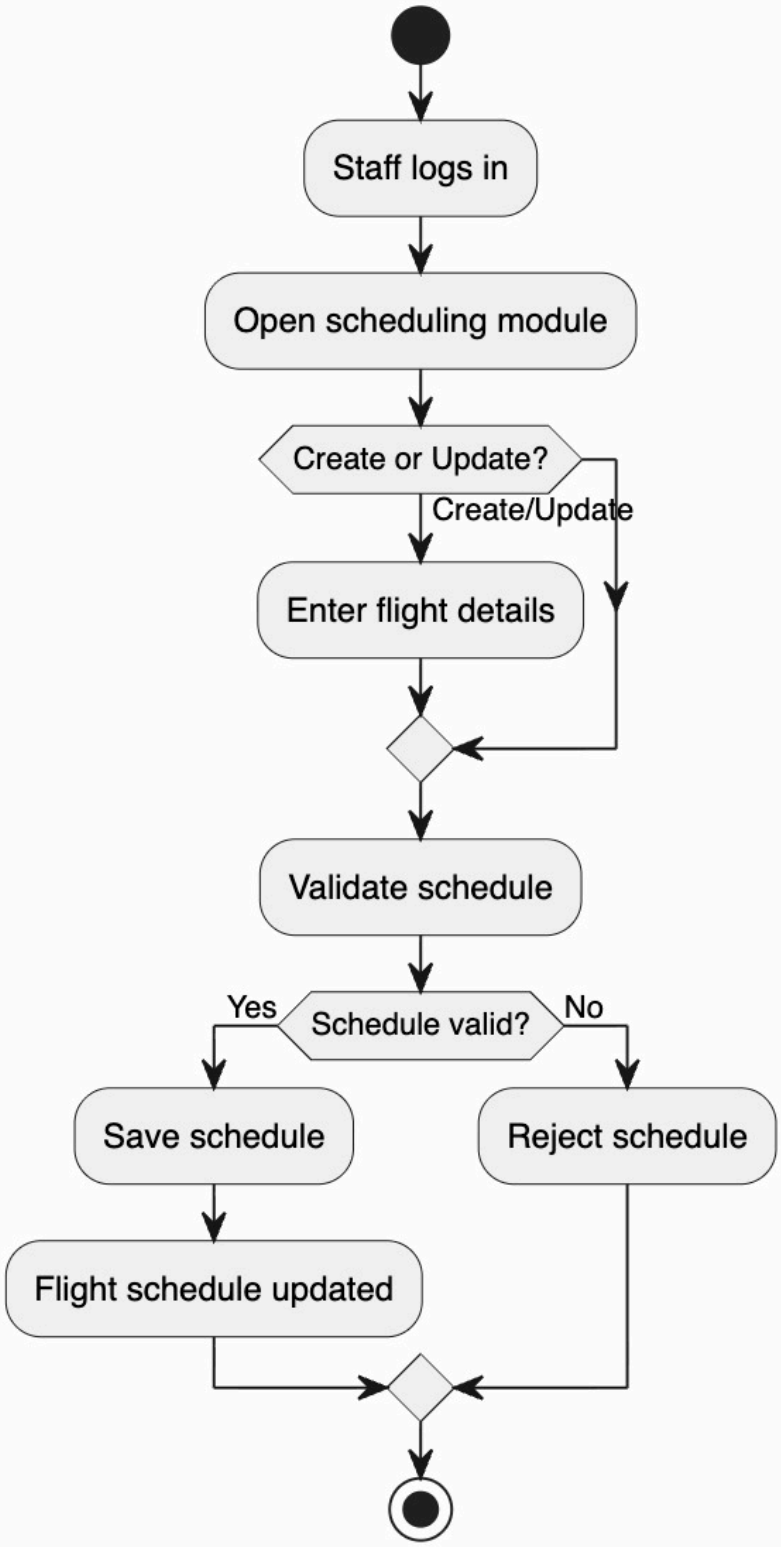
| UC Name | *UC-05: Assign Gates to Flights* |
|---|---|
| Summary | *Operations Staff assigns gates to scheduled flights.* |
| Dependency | *UC-06* |
| Actors | ***Primary:*** *Operations Staff* |
| Preconditions | ● *Flight schedule exists* |
| Description of the Main Sequence | ● *Step 1: View flight schedule*<br>● *Step 2: Select flight*<br>● *Step 3: Assign gate*<br>● *Step 4: Confirm assignment* |
| Description of the Alternative Sequence | ● *Step 1: Gate unavailable → system prompts another selection* |
| Non functional requirements | *Real-time updates*<br><br>*Conflict prevention* |
| Postconditions | *Gate assigned to flight* |

## Activity Diagram UC-05



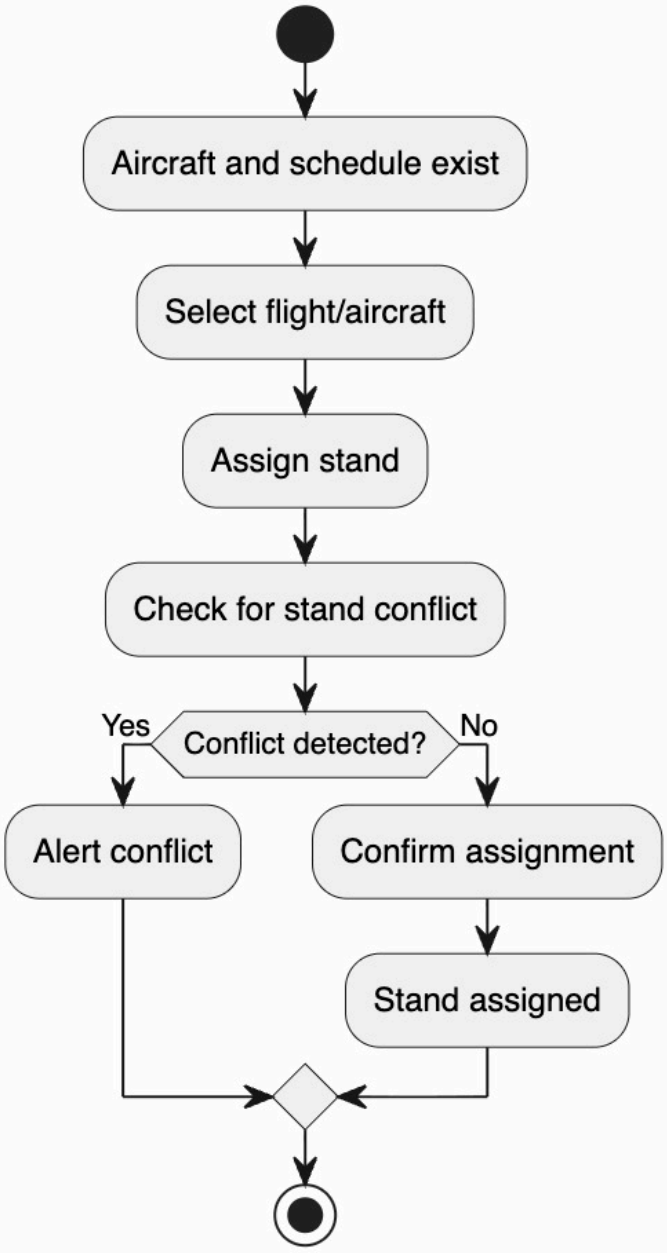
| UC Name | *UC-06: **Manage Flight Schedule*** |
|---|---|
| Summary | *Operations Staff creates and updates flight schedules.* |
| Dependency | *None* |
| Actors | ***Primary:*** *Operations Staff* |
| Preconditions | • *Staff logged in* |
| Description of the Main Sequence | • *Open scheduling module*<br>• *Create/update flight details*<br>• *Save schedule* |
| Description of the Alternative Sequence | • *Invalid schedule → rejected* |
| Non functional requirements | *High availability; Data accuracy* |
| Postconditions | *Flight schedule updated* |

## UC-06: Manage Flight Schedule - Activity Diagram



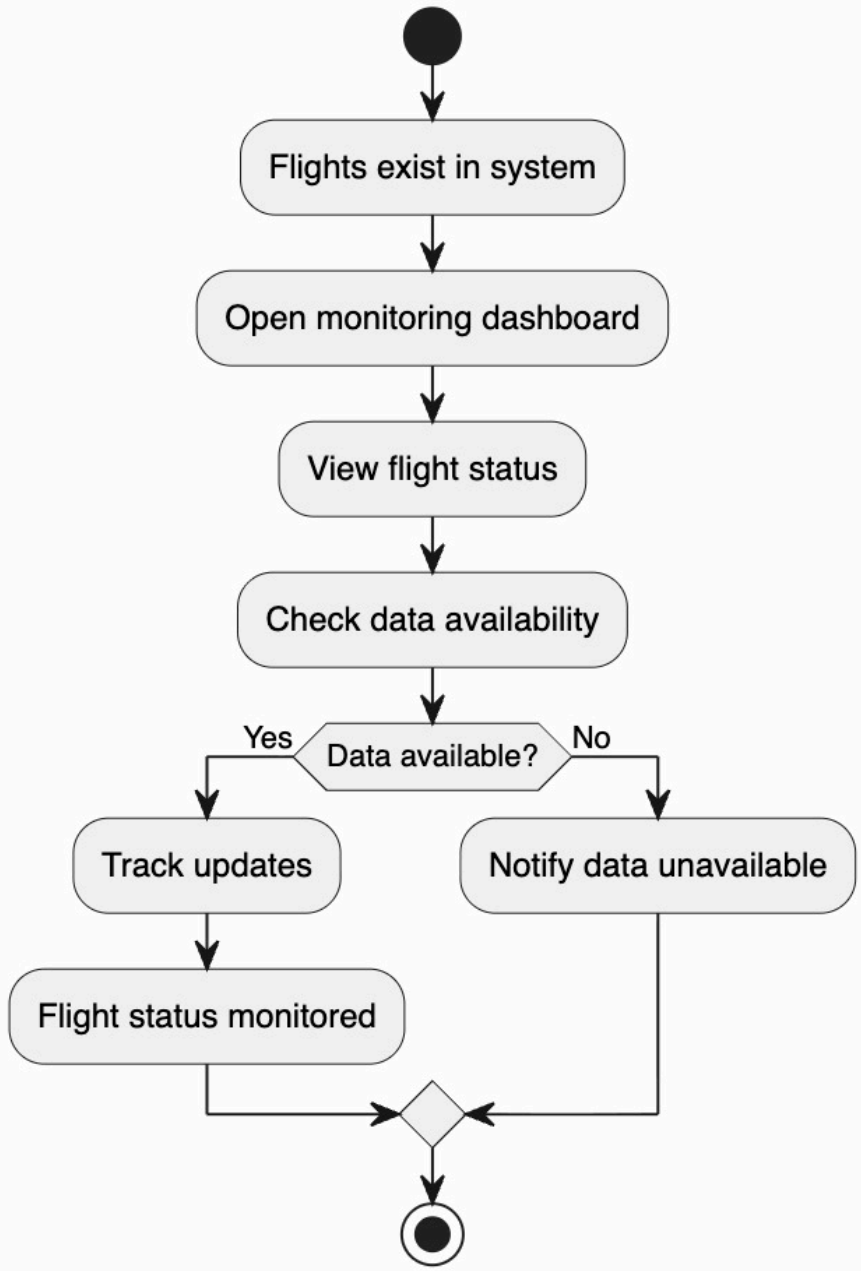
| UC Name | ***UC-07: Assign Aircraft Stands*** |
|---|---|
| Summary | *Operations Staff assigns stands to aircraft.* |
| Dependency | *UC-06* |
| Actors | *Operations Staff* |
| Preconditions | ● *Aircraft and schedule exist* |
| Description of the Main Sequence | ● *Select flight/aircraft*<br>● *Assign stand*<br>● *Confirm assignment* |
| Description of the Alternative Sequence | ● *Stand conflict → alert* |
| Non functional requirements | *Conflict detection; Real-time updates* |
| Postconditions | *Stand assigned* |

## UC-07: Assign Aircraft Stands - Activity Diagram



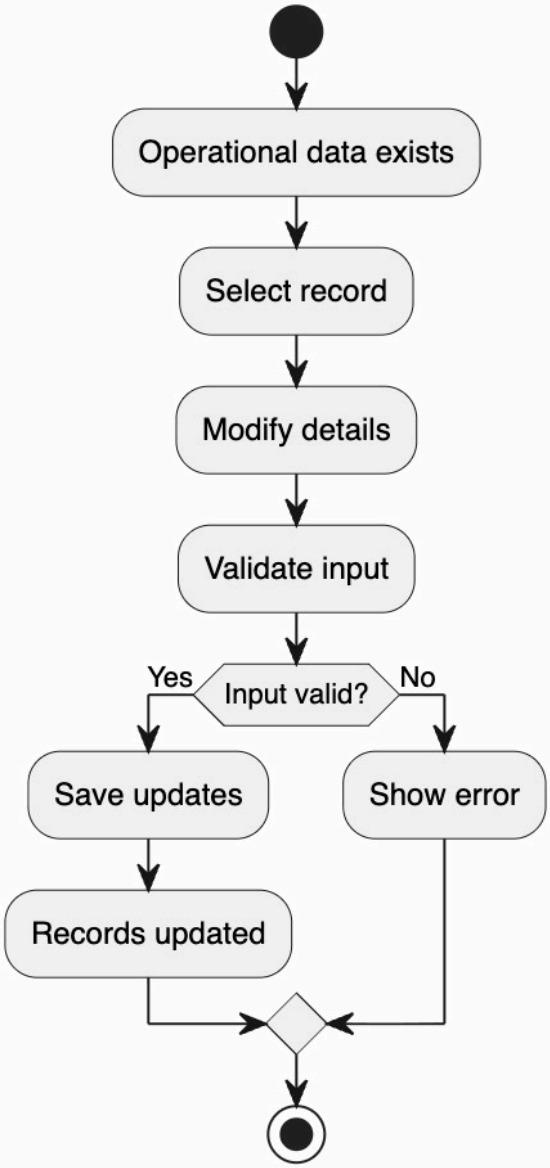
| UC Name | ***UC-08: Monitor Flight Status*** |
|---|---|
| | |
| Summary | *Operations Staff monitors flight status.* |
| Dependency | *UC-06* |
| Actors | *Operations Staff* |
| Preconditions | ● *Flights exist in system* |
| Description of the Main Sequence | ● *Open monitoring dashboard*<br>● *View flight status*<br>● *Track updates* |
| Description of the Alternative Sequence | ● *Data unavailable → notification* |
| Non functional requirements | *Real-time data; High performance* |
| Postconditions | *Flight status monitored* |

## UC-08: Monitor Flight Status - Activity Diagram



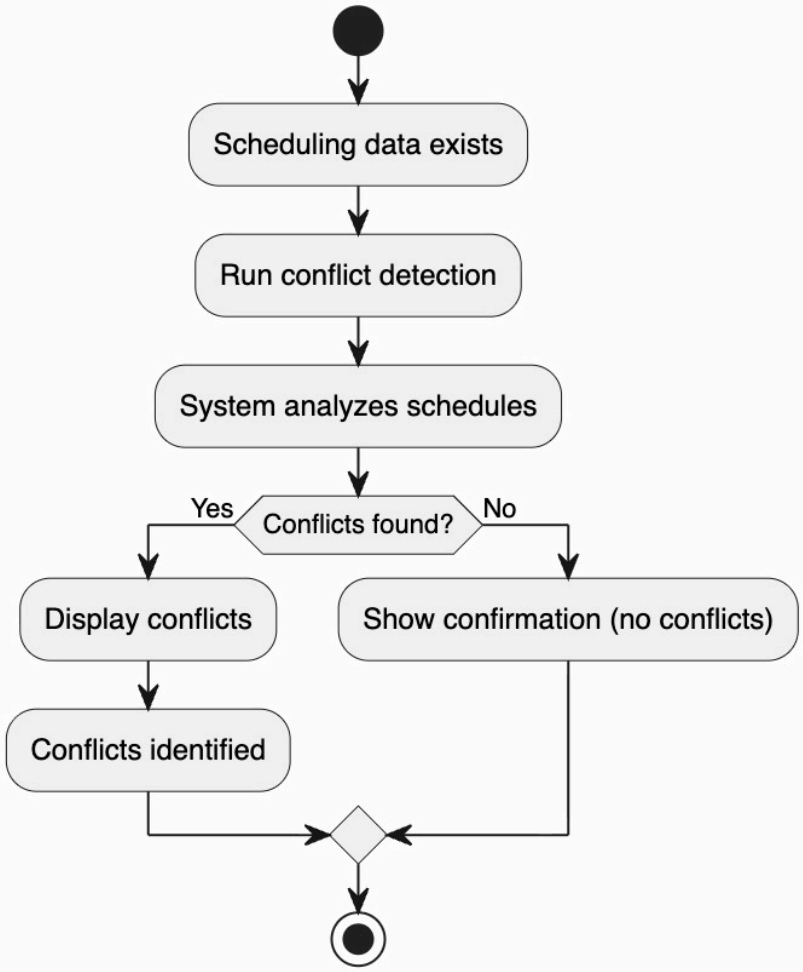
| UC Name | ***UC-09: Update Operational Records*** |
|---|---|
| | |
| Summary | *Operations Staff updates operational records.* |
| Dependency | *UC-05, UC-07* |
| Actors | *Operations Staff* |
| Preconditions | ● *Operational data exists* |
| Description of the Main Sequence | ● *Select record*<br>● *Modify details*<br>● *Save updates* |
| Description of the Alternative Sequence | ● *Invalid input → error* |
| Non functional requirements | *Data consistency; Audit logging* |
| Postconditions | *Records updated* |

## UC-09: Update Operational Records - Activity Diagram



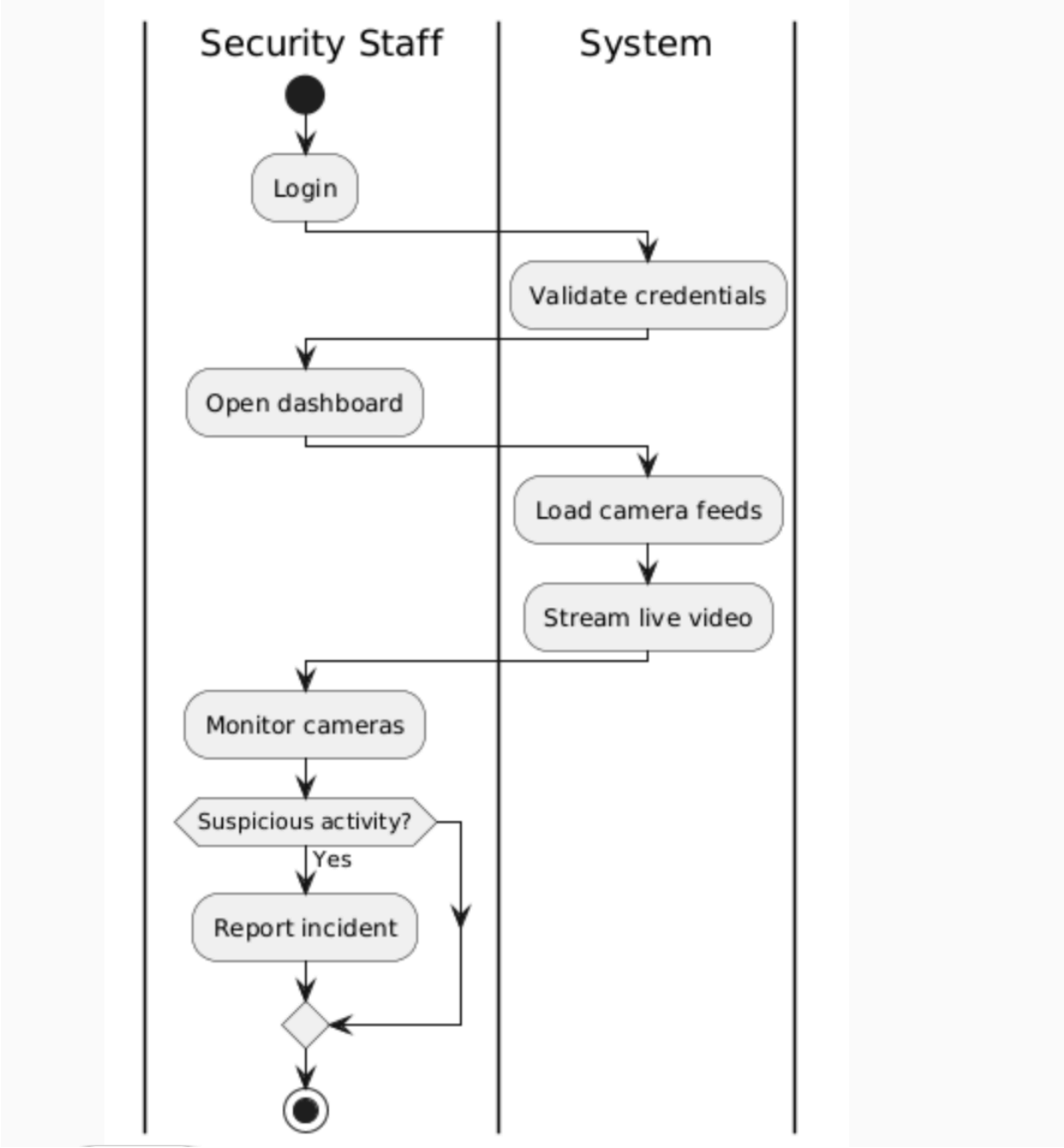
| UC Name | ***UC-10: Detect Scheduling Conflicts*** |
|---|---|
| | |
| Summary | *Operations Staff detects conflicts.* |
| Dependency | *UC-05, UC-07* |
| Actors | *Operations Staff* |
| Preconditions | ● *Scheduling data exists* |
| Description of the Main Sequence | ● *Run conflict detection*<br>● *System analyzes schedules*<br>● *Display conflicts* |
| Description of the Alternative Sequence | ● *No conflicts → confirmation* |
| Non functional requirements | *Fast processing; Accurate detection* |
| Postconditions | *Conflicts identified* |

## UC-10: Detect Scheduling Conflicts - Activity Diagram



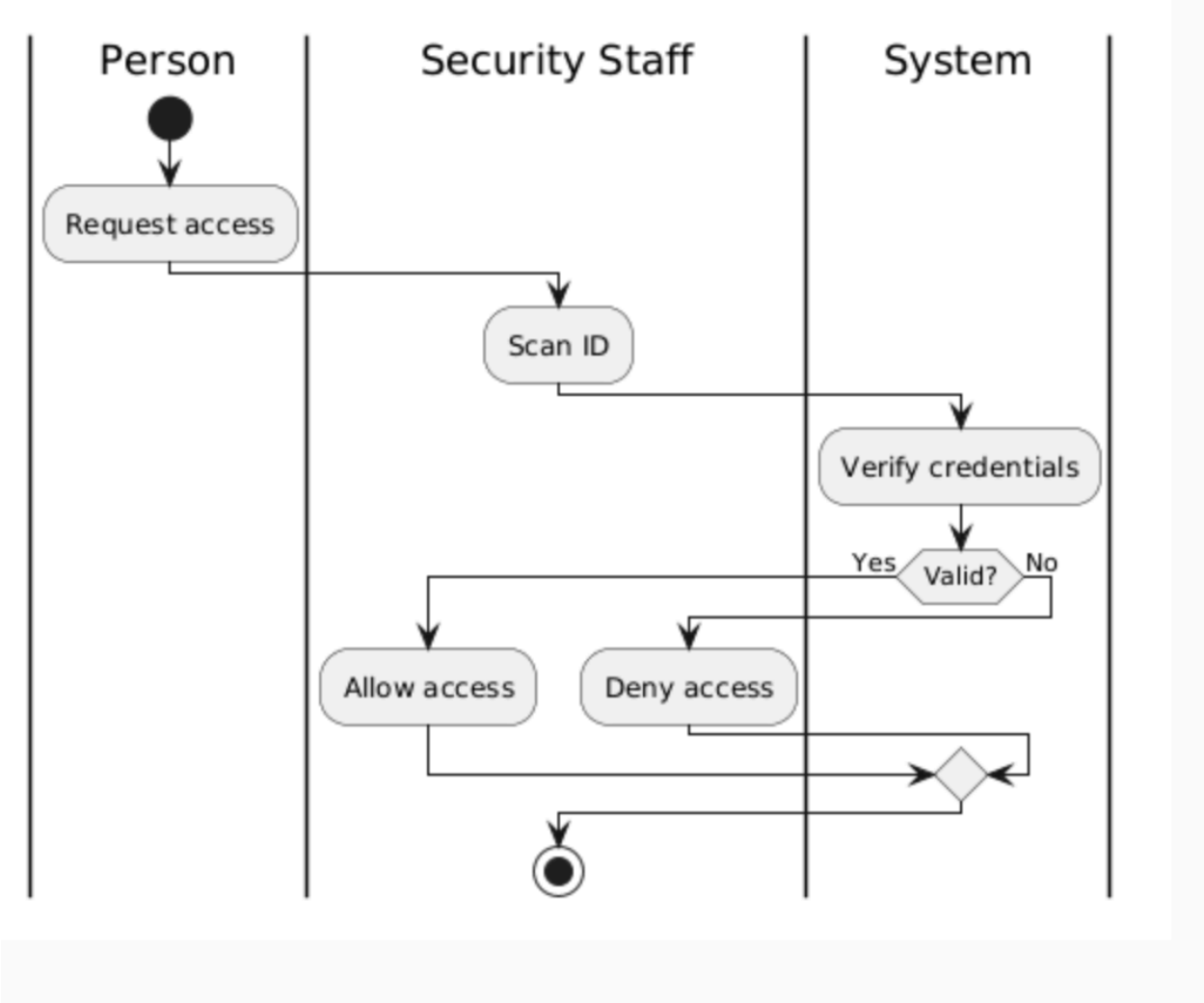
| UC Name | *UC-11:Monitor Surveillance Camera* |
|---|---|
| Summary | *Security staff monitors live camera feeds to detect suspicious activities and ensure airport safety.* |
| Dependency | *UC-01 and UC-02* |
| Actors | ***Primary:*** *Security Staff*<br>***Secondary:*** *Surveillance System* |
| Preconditions | ● *Security staff is logged into the system*<br>● *Cameras are active and connected* |
| Description of the Main Sequence | ● *Step 1: Security staff logs into the system*<br>● *Step 2: User opens surveillance monitoring dashboard*<br>● *Step 3: System displays live camera feeds*<br>● *Step 4: Security staff selects specific camera views*<br>● *Step 5: Security staff monitors activities in real time*<br>● *Step 6: Suspicious activity is identified* |
| Description of the Alternative Sequence | ● *Step 1: Camera feed is unavailable*<br>● *Step 2: System displays error message*<br>● *Step 3: Staff reports issue to maintenance* |
| Non functional requirements | *Real-time video streaming*<br>*High system availability (24/7)*<br>*Secure access to surveillance feeds* |

**Activity Diagram UC-11**



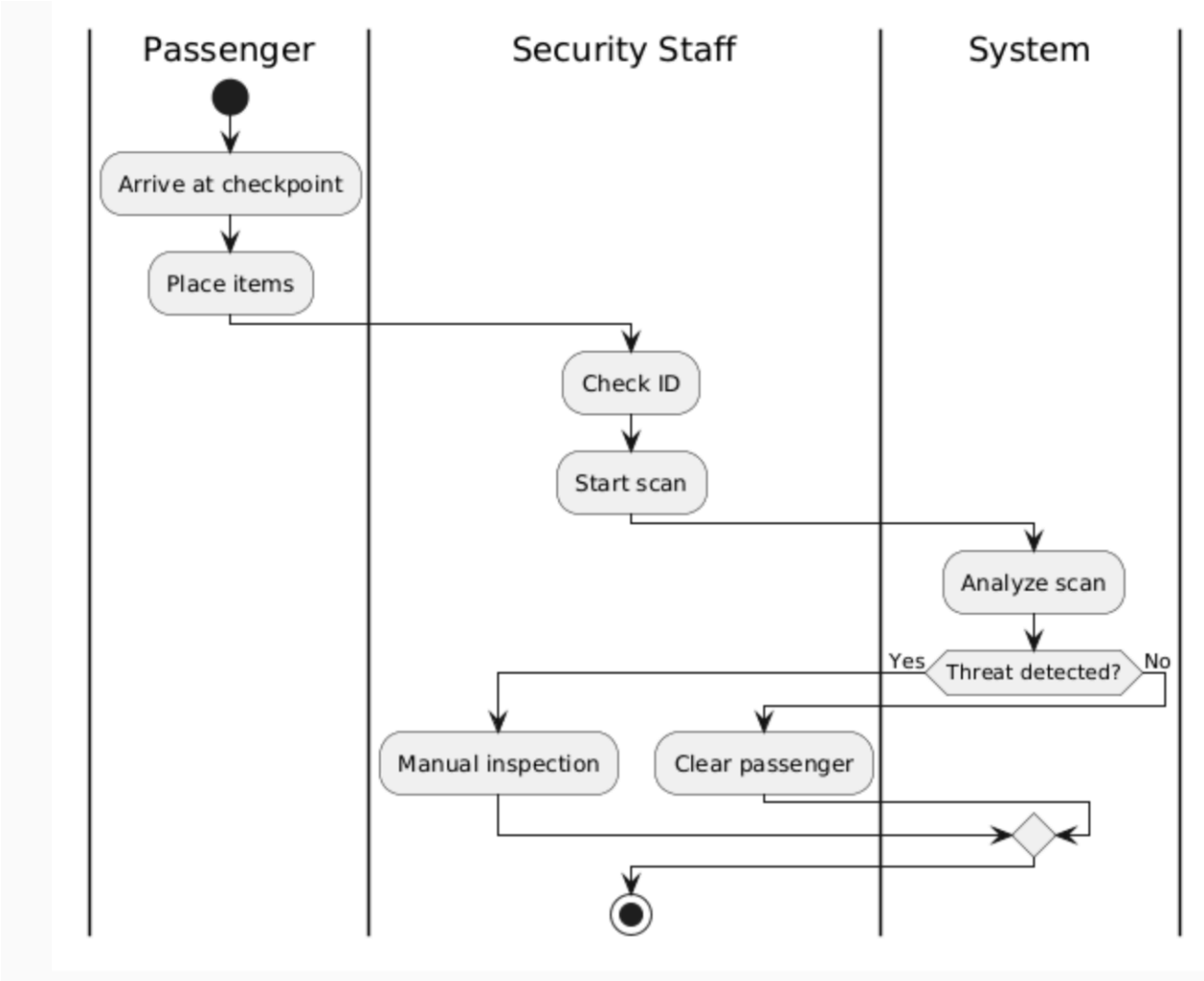
| UC Name | *UC-12 – Control Access to Restricted Areas* |
|---|---|
| Summary | *Security staff verifies and controls access to restricted airport zones.* |
| Dependency | *UC-01* |
| Actors | ***Primary Actor:*** *Security Staff*<br>***Secondary Actor:*** *Staff Member / Passenger* |
| Preconditions | ● *Security staff is logged in*<br>● *Access control system is active* |
| Description of the Main Sequence | ● *Step 1: Person requests access to restricted area*<br>● *Step 2: Security staff scans ID or access badge*<br>● *Step 3: System verifies credentials*<br>● *Step 4: System grants or denies access*<br>● *Step 5: Security staff allows or blocks entry* |
| Description of the Alternative Sequence | ● *Step 1: Invalid or expired credentials detected*<br>● *Step 2: System denies access*<br>● *Step 3: Security staff reports incident* |
| Non functional requirements | *High accuracy detection*<br>*Fast processing*<br>*Compliance with safety regulations* |
| Postconditions | *Only authorized individuals gain access*<br>*Access attempt is logged* |

**Activity Diagram UC-12**



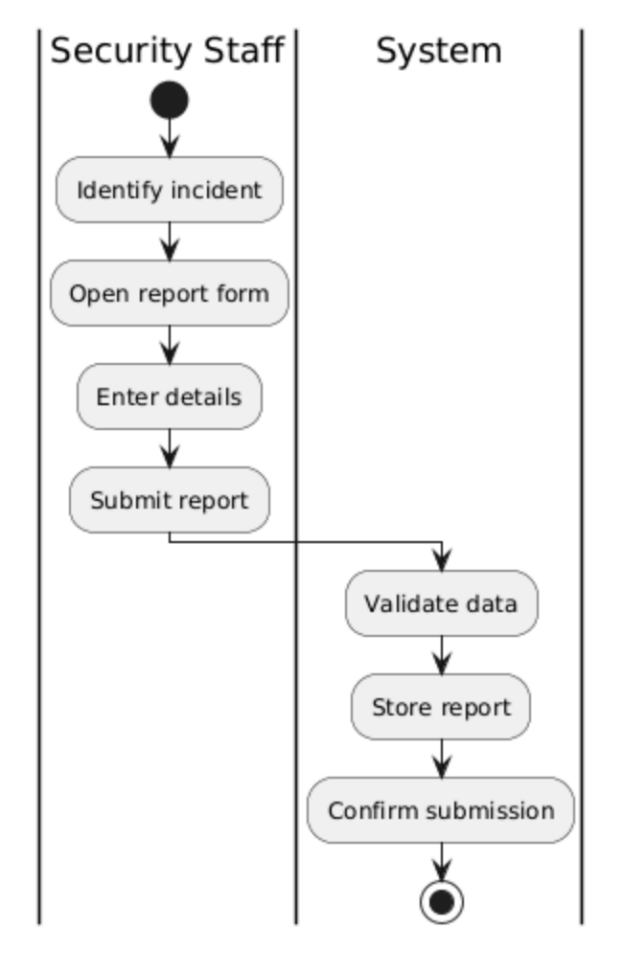
| UC Name | *UC-13 – Perform Security Screening* |
|---|---|
| Summary | *Security staff screens passengers and staff before entering secure areas.* |
| Dependency | *None* |
| Actors | ***Primary Actor:*** *Security Staff*<br>***Secondary Actor:*** *Passenger / Staff* |
| Preconditions | ● *Screening equipment is operational*<br>● *Passenger/staff is present* |
| Description of the Main Sequence | ● *Step 1: Passenger arrives at screening checkpoint*<br>● *Step 2: Security staff checks identification*<br>● *Step 3: Passenger places belongings on scanner*<br>● *Step 4: Security staff monitors scanning process*<br>● *Step 5: Screening results are displayed*<br>● *Step 6: Passenger is cleared* |
| Description of the Alternative Sequence | ● *Step 1: Suspicious item detected*<br>● *Step 2: Security staff performs manual inspection*<br>● *Step 3: Item is confiscated or escalated* |
| Non functional requirements | *Fast response time*<br>*Secure authentication*<br>*High reliability* |
| Postconditions | *Passenger is either cleared or flagged*<br>*Screening result is recorded* |

## UML Activity Diagram



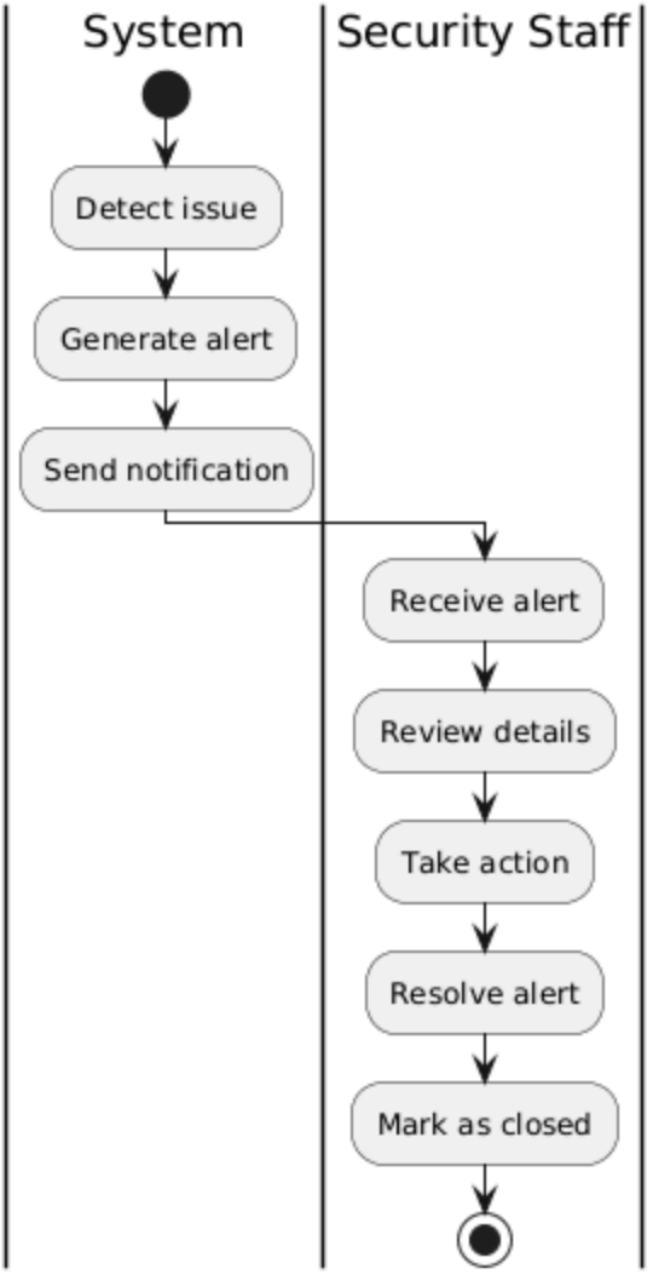
| UC Name | *UC-14– Record and Report Security Incident* |
|---|---|
| Summary | *Security staff records and reports incidents for investigation and auditing.* |
| Dependency | *UC-11* |
| Actors | ***Primary Actor:*** *Security Staff*<br>***Secondary Actor:*** *System Administrator* |
| Preconditions | ● *Security staff is logged in*<br>● *Incident has occurred* |
| Description of the Main Sequence | ● *Step 1: Security staff identifies incident*<br>● *Step 2: Staff opens incident reporting module*<br>● *Step 3: Staff enters incident details*<br>● *Step 4: Staff submits report*<br>● *Step 5: System stores report in database* |
| Description of the Alternative Sequence | ● *Step 1: Incomplete information entered*<br>● *Step 2: System prompts user to complete fields* |
| Non functional requirements | *Data integrity*<br>*Secure storage of reports*<br>*Audit logging* |
| Postconditions | *Incident is recorded in system*<br>*Report is available for review* |

## Activity Diagram UC-14



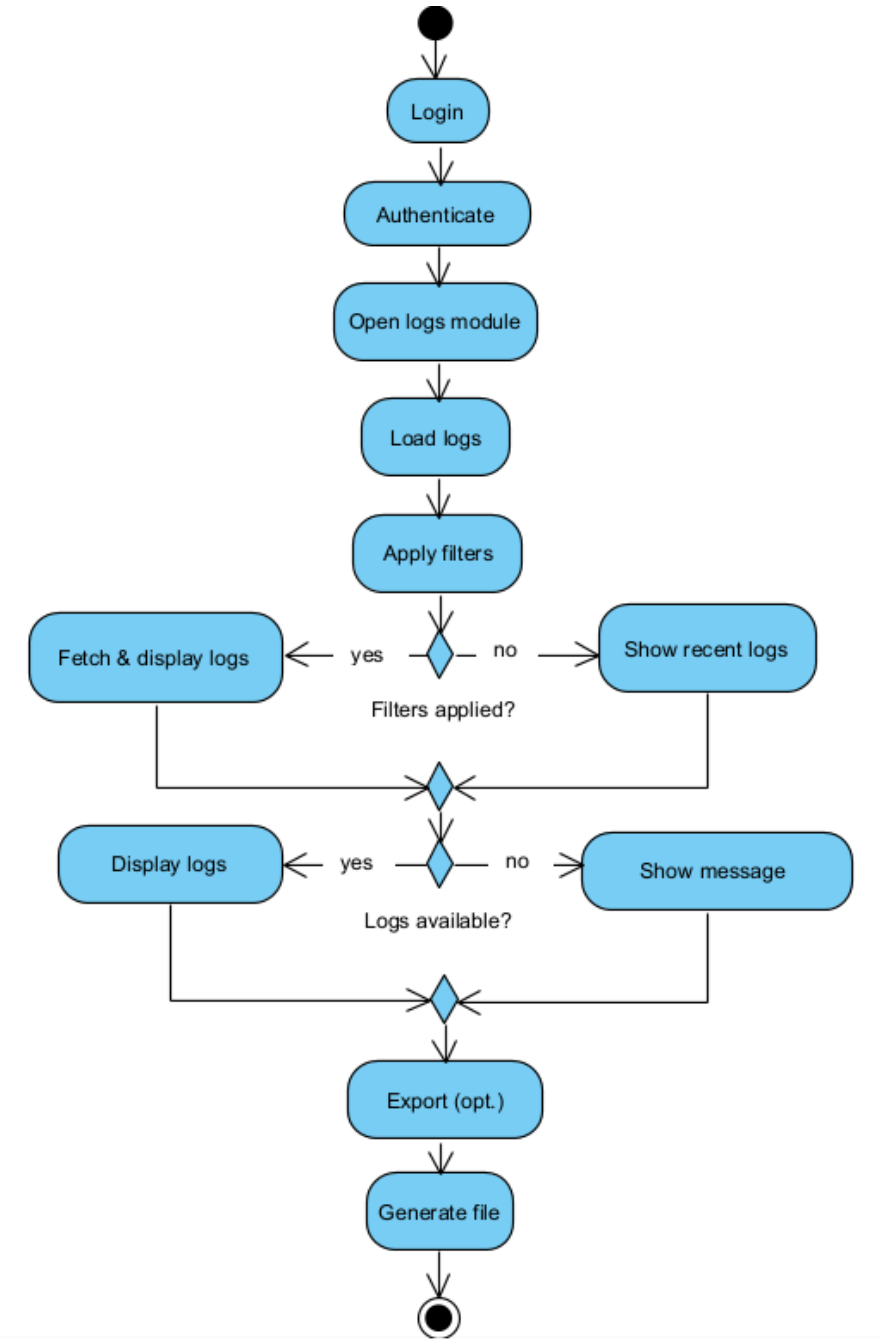
| UC Name | *UC-15– Respond to Security Alerts* |
|---|---|
| Summary | *Security staff receives and responds to alerts generated by the system.* |
| Dependency | *UC-11 and UC-12* |
| Actors | ***Primary Actor:*** *Security Staff*<br>***Secondary Actor:*** *System* |
| Preconditions | ● *System is operational*<br>● *Alert mechanism is active* |
| Description of the Main Sequence | ● *Step 1: System detects abnormal activity*<br>● *Step 2: System generates alert*<br>● *Step 3: Security staff receives notification*<br>● *Step 4: Staff reviews alert details*<br>● *Step 5: Staff takes appropriate action*<br>● *Step 6: Staff marks alert as resolved* |
| Description of the Alternative Sequence | ● *Step 1: False alert detected*<br>● *Step 2: Staff dismisses alert* |
| Non functional requirements | *Real-time alert delivery*<br>*High reliability*<br>*Secure communication* |
| Postconditions | *Alert is handled and resolved*<br>*Action is logged in the system* |

**Activity Diagram UC-15**



| UC Name | *UC-16: Review Security Logs and Reports* |
|---|---|
| Summary | *Security staff review security logs and reports to monitor risks and improve safety procedures.* |
| Dependency | *UC-03, UC-14, UC-15* |
| Actors | ***Primary:*** *Security Staff* |
| Preconditions | ● *Security staff is authenticated*<br>● *Security logs exist in the system* |
| Description of the Main Sequence | ● *Step 1: Security staff logs into the system*<br>● *Step 2: Navigates to security logs and reports module*<br>● *Step 3: Applies filters (date, type, location)*<br>● *Step 4: Views logs and reports*<br>● *Step 5: Optionally exports reports* |
| Description of the Alternative Sequence | ● *Step 1: No filters applied → system shows recent logs*<br>● *Step 2: No data found → system displays message* |
| Non functional requirements | ● *Data confidentiality and integrity*<br>● *Fast query response time*<br>● *Secure access control* |
| Postconditions | *Security logs and reports are reviewed* |

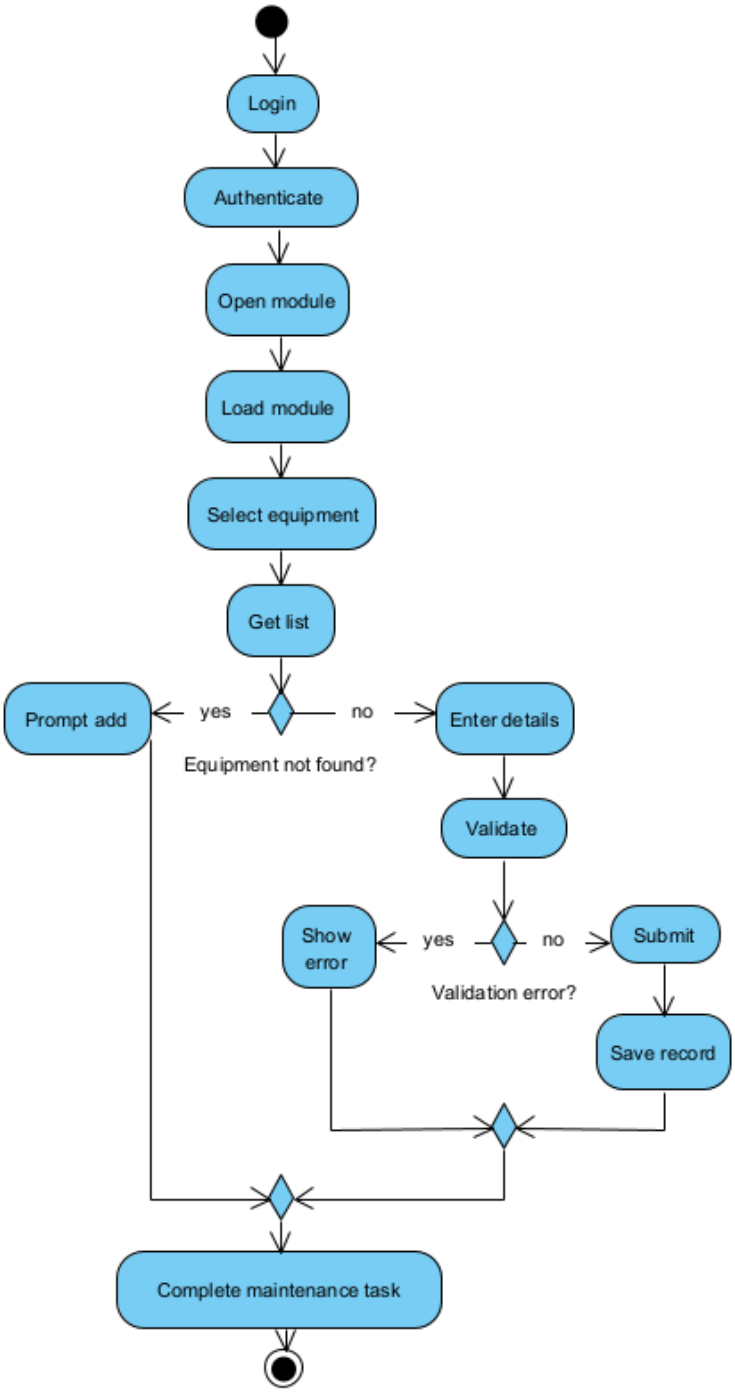
## Activity Diagramm. UC-16




| UC Name | *UC-17: Record Maintenance Activities* |
|---|---|

| | |
|---|---|
| Summary | *Maintenance staff record completed maintenance activities for equipment.* |
| Dependency | *UC-21* |
| Actors | ***Primary:*** *Maintenance Staff* |
| Preconditions | ● *Maintenance staff is authenticated*<br>● *Equipment exists in the system* |
| Description of the Main Sequence | ● *Step 1: Maintenance staff logs into the system*<br>● *Step 2: Navigates to maintenance logging module*<br>● *Step 3: Selects equipment*<br>● *Step 4: Enters maintenance details*<br>● *Step 5: Submits the record* |
| Description of the Alternative Sequence | ● *Step 1: Equipment not found → system prompts to add equipment*<br>● *Step 2: Missing required fields → system shows error* |
| Non functional requirements | ● *Data accuracy*<br>● *System reliability*<br>● *Quick data entry response* |
| Postconditions | *Maintenance activity is recorded* |

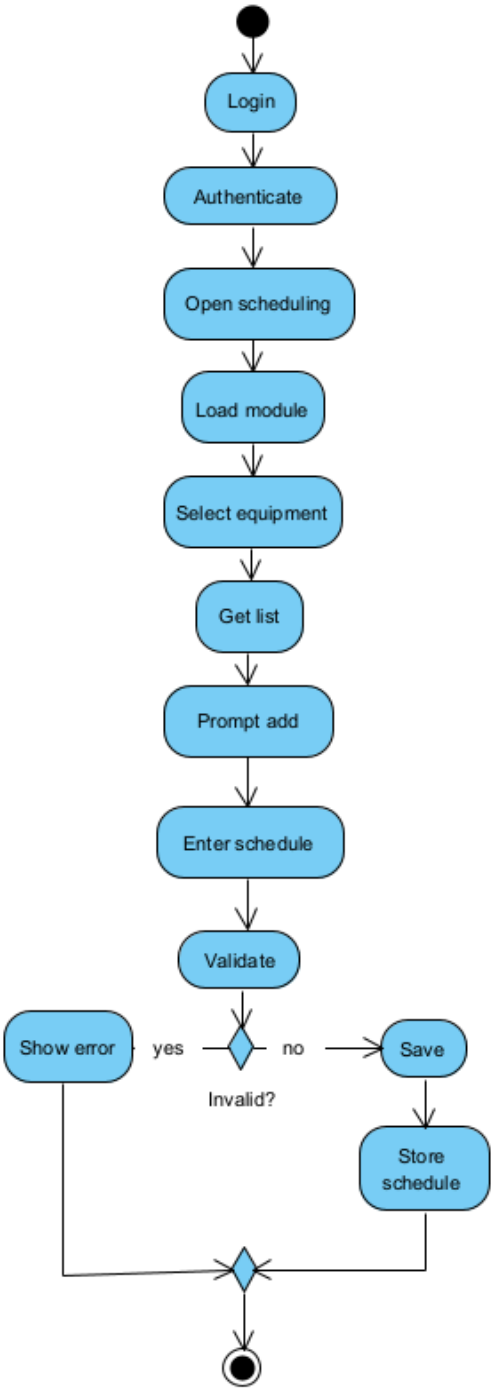
## Activity Diagram UC-17




| UC Name | *UC-18: Schedule Preventive Maintenance* |
|---|---|

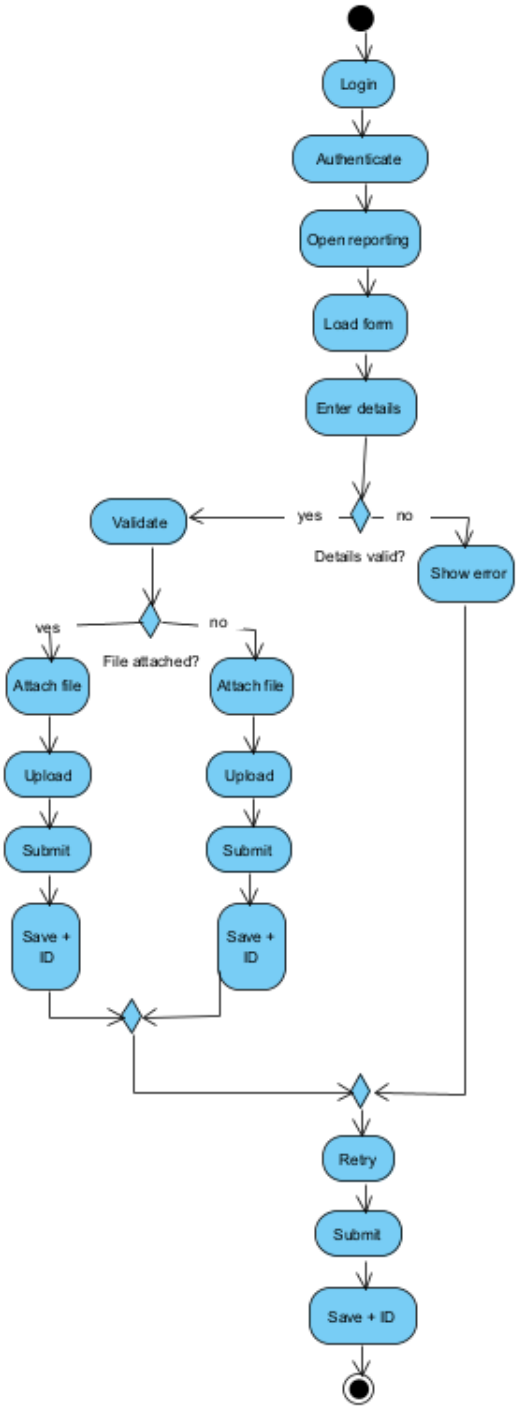
| | |
|---|---|
| Summary | *Maintenance staff schedule preventive maintenance tasks to avoid equipment failure.* |
| Dependency | *UC-21* |
| Actors | ***Primary:*** *Maintenance Staff* |
| Preconditions | ● *Maintenance staff is authenticated*<br>● *Equipment list is available* |
| Description of the Main Sequence | ● *Step 1: Maintenance staff logs into the system*<br>● *Step 2: Navigates to preventive maintenance module*<br>● *Step 3: Selects equipment*<br>● *Step 4: Defines schedule (date, frequency)*<br>● *Step 5: Saves schedule* |
| Description of the Alternative Sequence | ● *Step 1: Invalid date entered → system shows error*<br>● *Step 2: Equipment not available → system prompts registration* |
| Non functional requirements | ● *Scheduling accuracy*<br>● *System availability*<br>● *Notification support* |
| Postconditions | *Preventive maintenance schedule is created* |

## Activity Diagramm. UC-18


Login
Authenticate
Open scheduling
Load module
Select equipment
Get list
Prompt add
Enter schedule
Validate
Show error
yes
no
Invalid?
Save
Store schedule

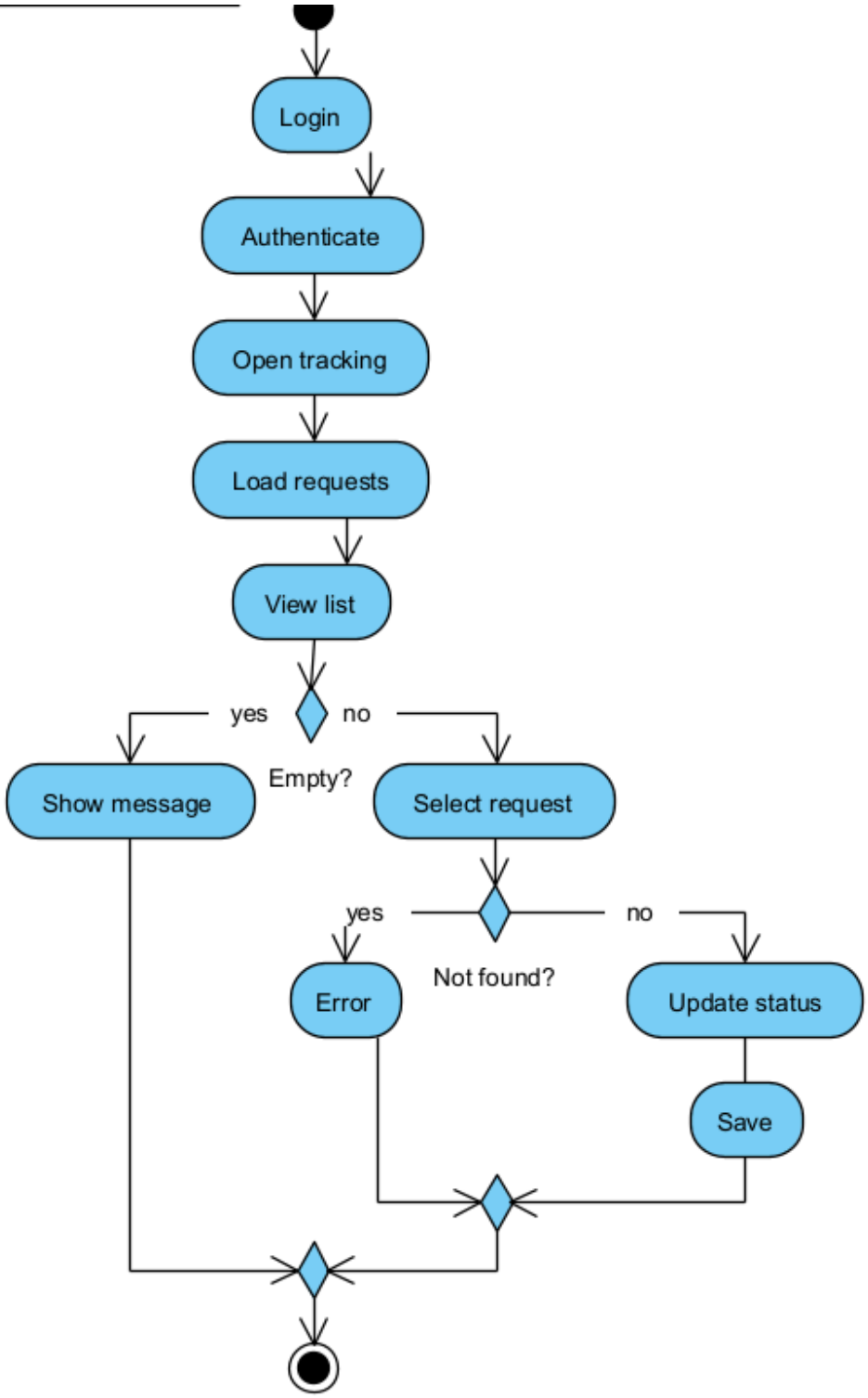
| UC Name | *UC-19: Report Infrastructure Issues* |
|---|---|
| Summary | *Maintenance staff report infrastructure or equipment issues.* |
| Dependency | *None* |
| Actors | ***Primary:*** *Maintenance Staff* |
| Preconditions | ● *Maintenance staff is authenticated* |
| Description of the Main Sequence | ● *Step 1: Maintenance staff logs into the system*<br>● *Step 2: Navigates to issue reporting module*<br>● *Step 3: Enters issue details*<br>● *Step 4: Attaches supporting files (optional)*<br>● *Step 5: Submits the report* |
| Description of the Alternative Sequence | ● *Step 1: Missing required information → system shows error*<br>● *Step 2: File upload fails → system prompts retry* |
| Non functional requirements | ● *Data integrity*<br>● *File upload reliability*<br>● *System usability* |
| Postconditions | *Issue report is submitted and stored* |

## Activity Diagramm. UC-19



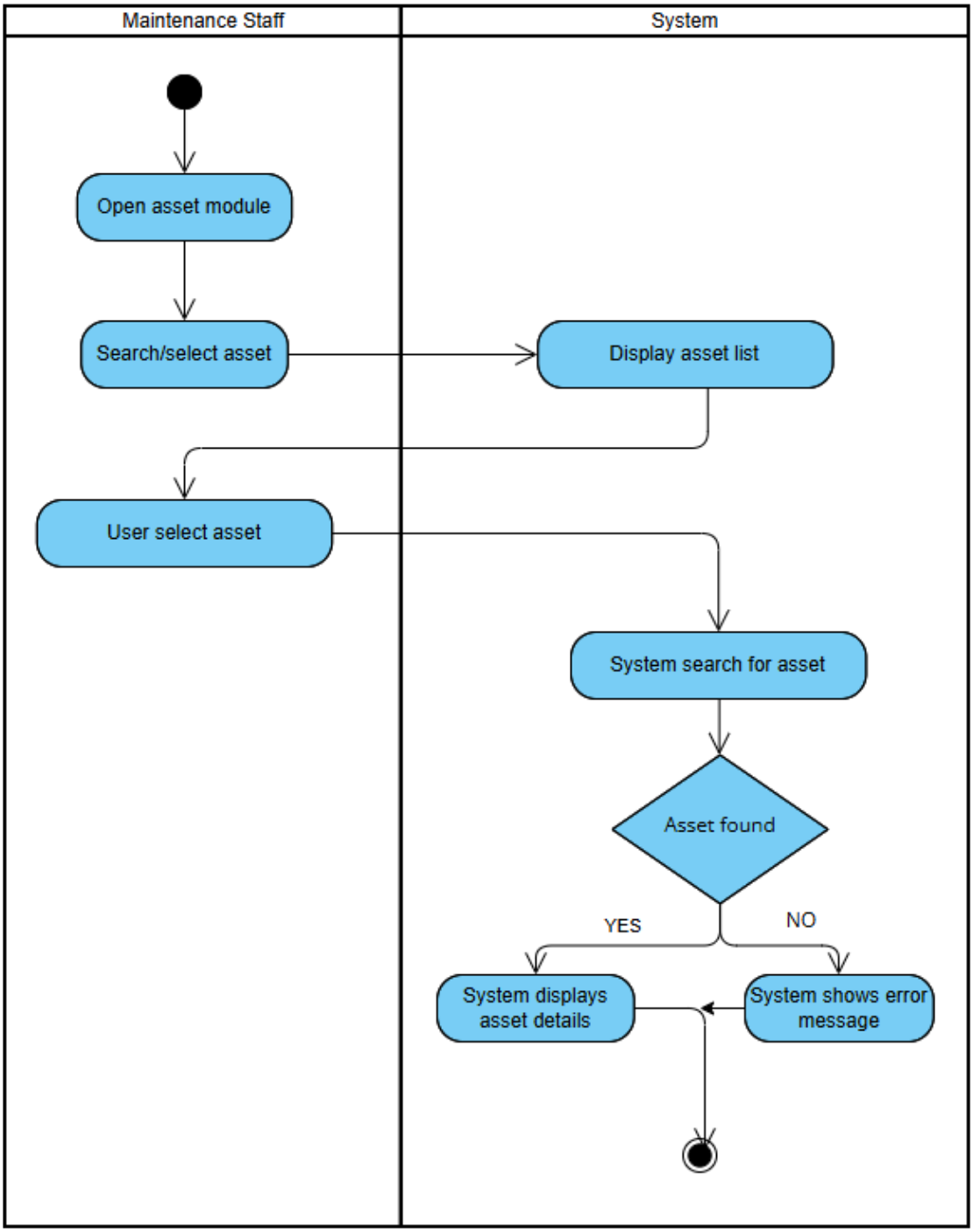
| UC Name | *UC-20: Track Maintenance Requests* |
|---|---|
| Summary | *Maintenance staff track and monitor maintenance request status.* |
| Dependency | *UC-19, UC-17* |
| Actors | ***Primary:*** *Maintenance Staff* |
| Preconditions | ● *Maintenance staff is authenticated*<br>● *Maintenance requests exist* |
| Description of the Main Sequence | ● *Step 1: Maintenance staff logs into the system*<br>● *Step 2: Navigates to maintenance tracking module*<br>● *Step 3: Views list of requests*<br>● *Step 4: Selects a request*<br>● *Step 5: Views or updates status* |
| Description of the Alternative Sequence | ● *Step 1: No requests available → system shows empty list*<br>● *Step 2: Request not found → system shows error* |
| Non functional requirements | ● *Real-time updates*<br>● *System performance*<br>● *Data consistency* |
| Postconditions | *Maintenance request status is viewed or updated* |

## Activity Diagramm. UC-20



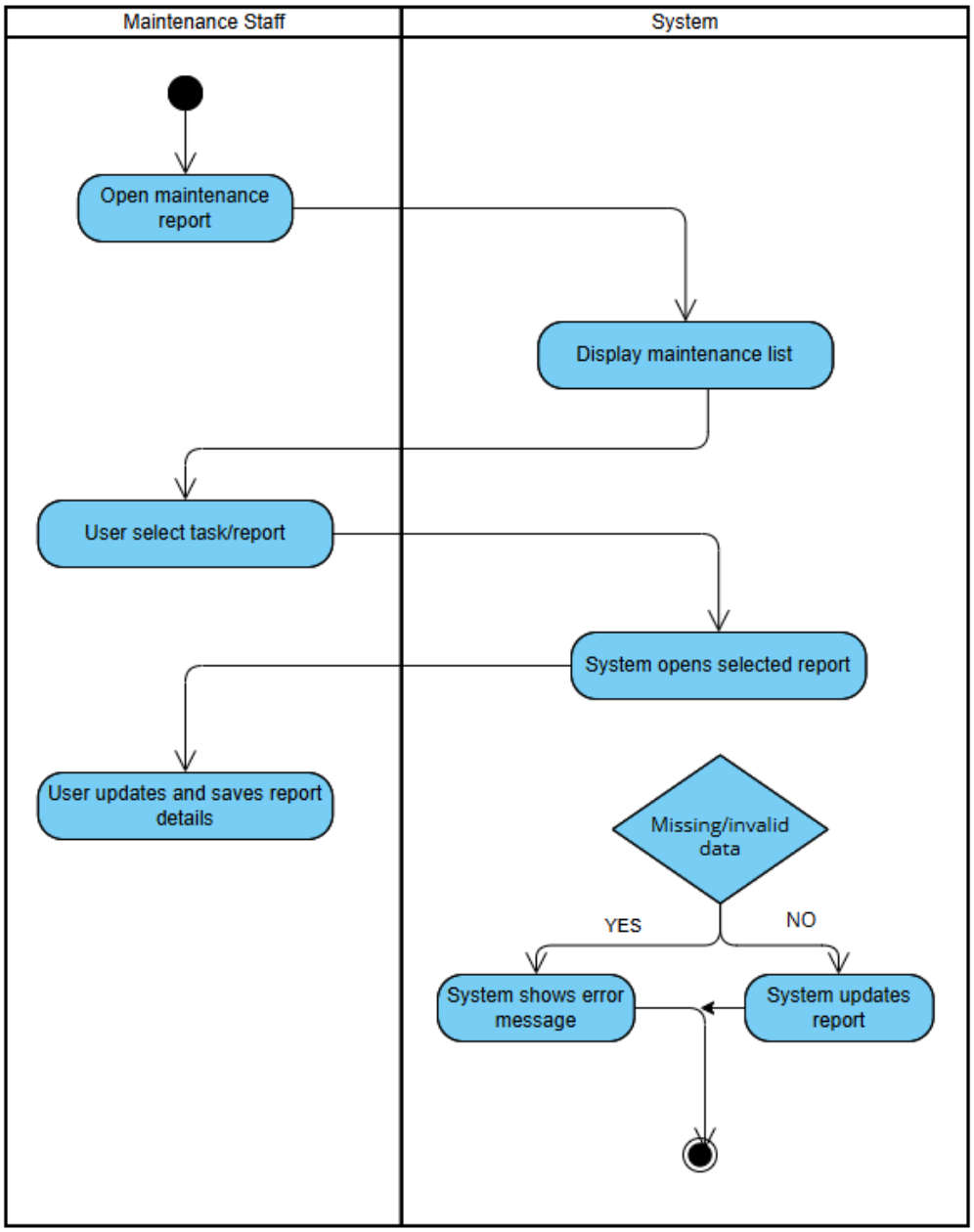
| UC Name | *UC-21 — Access Asset Information* |
|---|---|
| Summary | • Maintenance staff accesses asset information for airport equipment. |
| Dependency | *None* |
| Actors | ***Primary:*** *Maintenance Staff* |
| Preconditions | • *Asset records exist* |
| Description of the Main Sequence | • *Step 1: View asset list*<br>• *Step 2: Select asset*<br>• *Step 3: View asset details* |
| Description of the Alternative Sequence | • *Step 1: Asset not found → system shows error* |
| Non functional requirements | • *Fast access*<br>• *Data accuracy* |
| Postconditions | *Asset information displayed* |

## UML-Activity Diagram



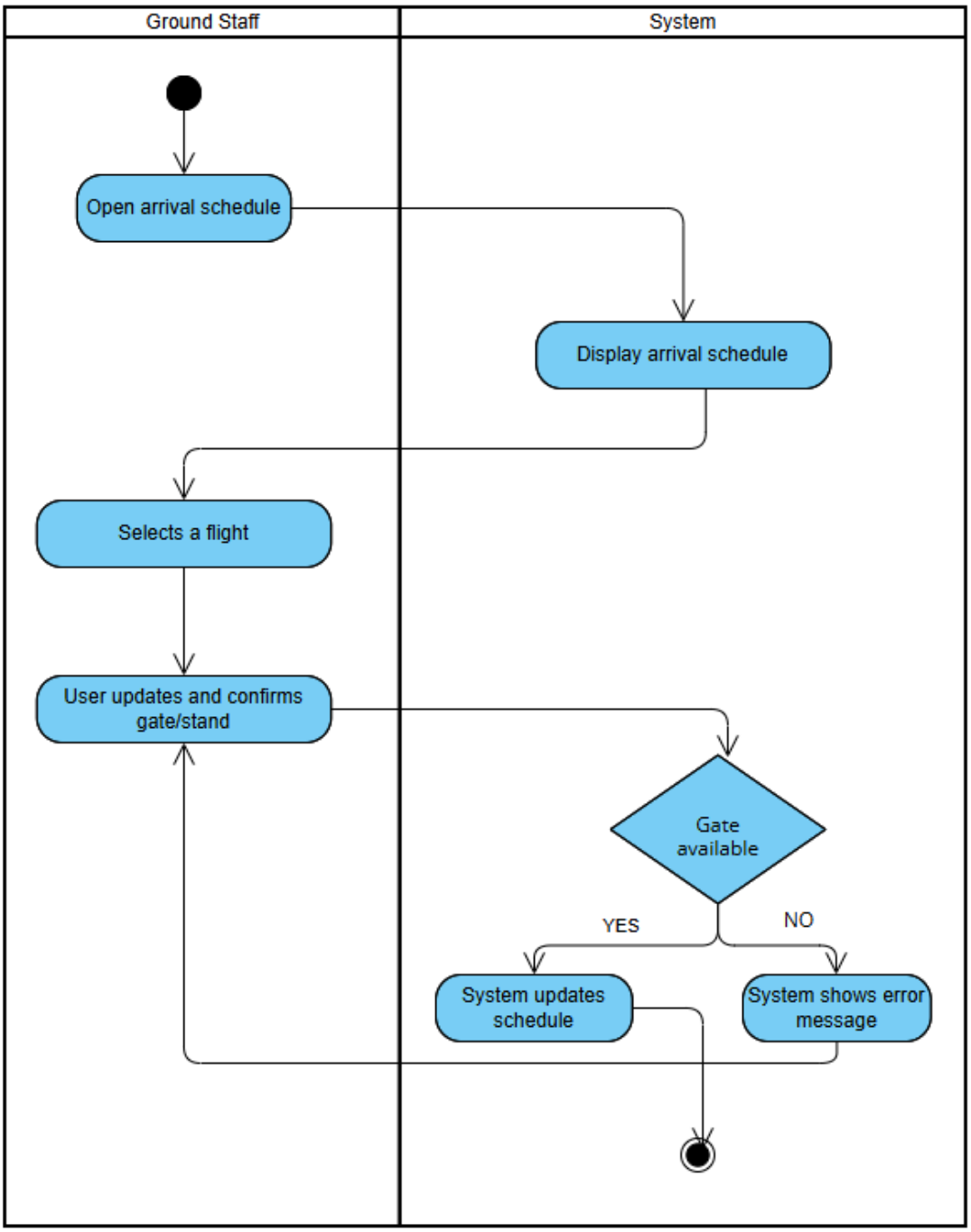
| UC Name | *UC-22 — Update Maintenance Report* |
|---|---|
| Summary | • Maintenance staff repair and maintenance reports |
| Dependency | *UC-21* |
| Actors | ***Primary:*** *Maintenance Staff* |
| Preconditions | • *Maintenance records exist* |
| Description of the Main Sequence | • *Step 1: View maintenance tasks*<br>• *Step 2: Select task*<br>• *Step 3: Update reports*<br>• *Step 4: Save changes* |
| Description of the Alternative Sequence | • *Step 1: Missing Data → system shows error* |
| Non functional requirements | • *Reliable saving*<br>• *Data consistency* |
| Postconditions | *Report Updated* |

## UML-Activity Diagram



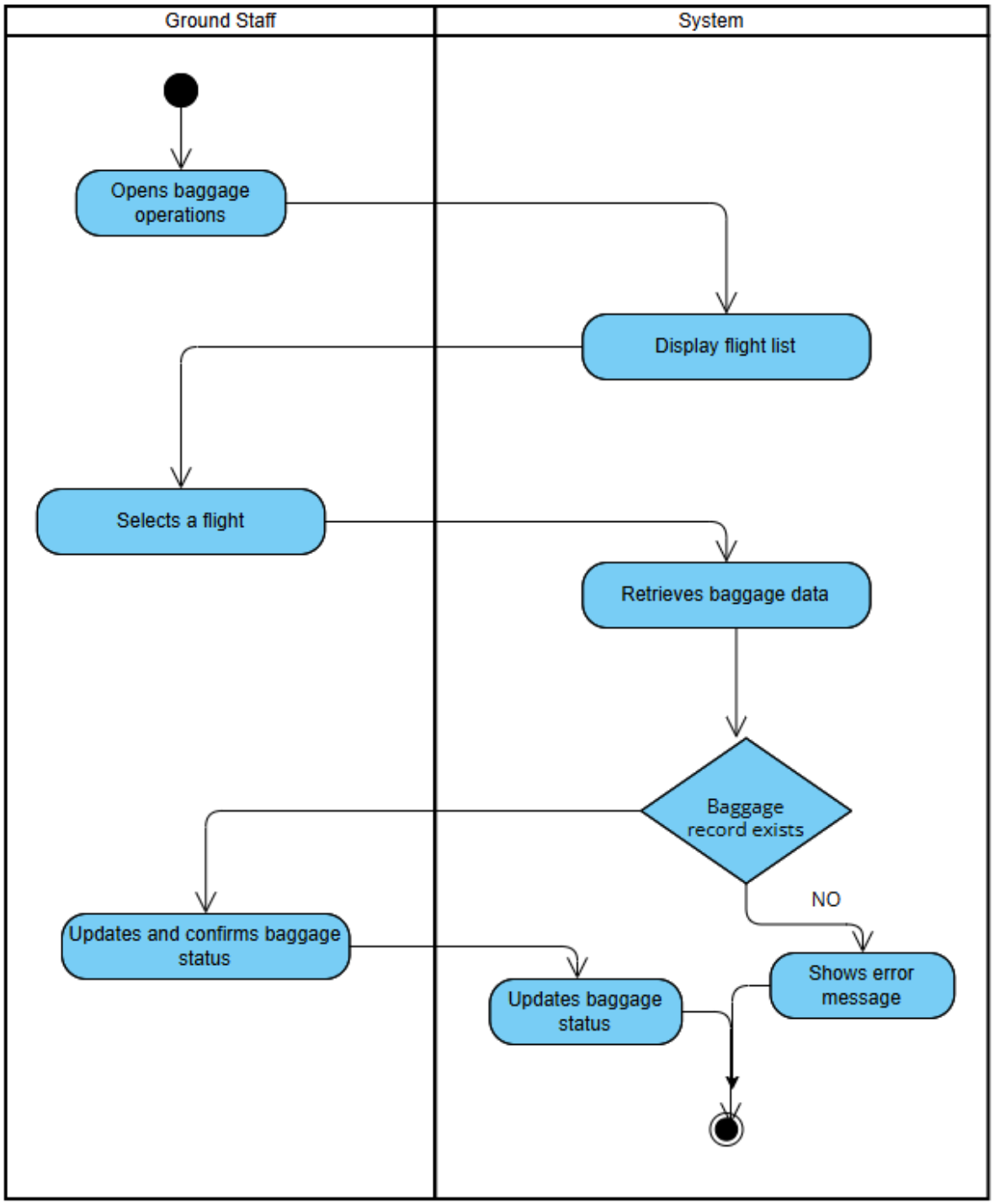
| UC Name | *UC-23 — Manage Arrival Schedule* |
|---|---|
| Summary | • Ground staff manages aircraft arrival schedules. |
| Dependency | *None* |
| Actors | ***Primary:*** *Ground Handling Staff* |
| Preconditions | • *Flight schedule exists* |
| Description of the Main Sequence | • *Step 1: View arrival schedule*<br>• *Step 2: Select flight*<br>• *Step 3: Update gate/stand*<br>• *Step 4: Confirm update* |
| Description of the Alternative Sequence | • *Step 1: Gate unavailable → select another* |
| Non functional requirements | • *Real-time updates*<br>• *Conflict prevention* |
| Postconditions | *Schedule updated* |

## UML-Activity Diagram



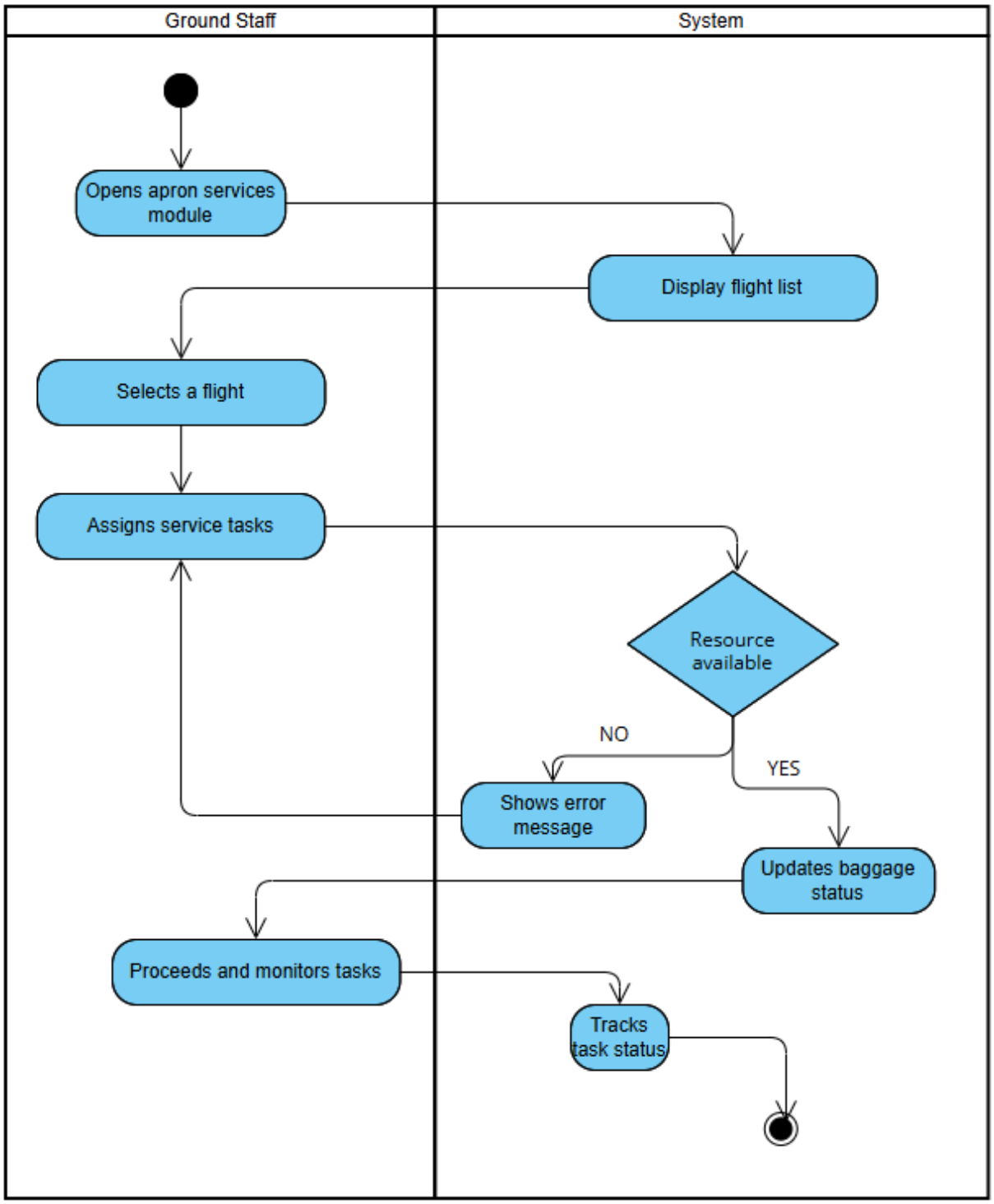
| UC Name | *UC-24 — Manage Baggage Operations* |
|---|---|
| Summary | • Ground staff manages baggage loading and unloading operations. |
| Dependency | *UC-23* |
| Actors | ***Primary:*** *Ground Handling Staff* |
| Preconditions | • *Flight and baggage status* |
| Description of the Main Sequence | • *Step 1: View flight list*<br>• *Step 2: Select flight*<br>• *Step 3: Update baggage status*<br>• *Step 4: Confirm update* |
| Description of the Alternative Sequence | • *Step 1: No baggage record → system shows error* |
| Non functional requirements | • *Real-time tracking*<br>• *Accuracy* |
| Postconditions | *Baggage status updated* |

## UML-Activity Diagram



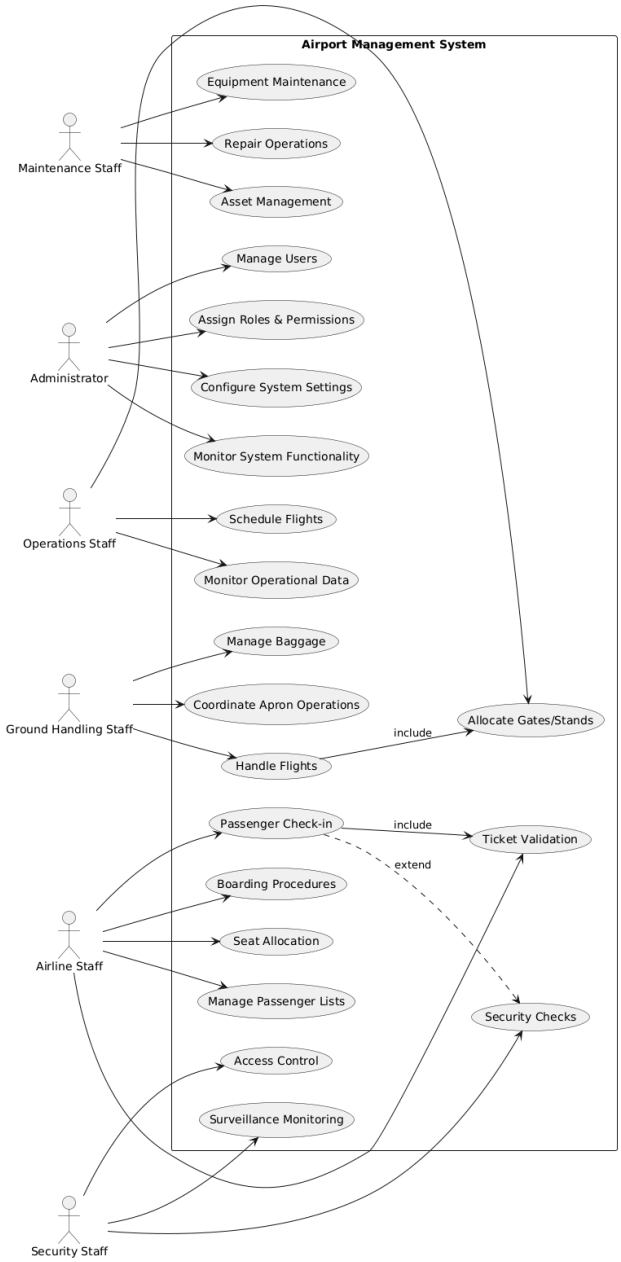
| UC Name | *UC-25 — Manage Apron Services* |
|---|---|
| Summary | • Ground staff assigns and monitors apron service tasks. |
| Dependency | *UC-23* |
| Actors | ***Primary:*** *Ground Handling Staff* |
| Preconditions | • *Flight schedule exists* |
| Description of the Main Sequence | • *Step 1: View flights*<br>• *Step 2: Select flight*<br>• *Step 3: Assign service tasks*<br>• *Step 4: Monitor tasks* |
| Description of the Alternative Sequence | • *Step 1: Resource unavailable → choose another* |
| Non functional requirements | • *Real-time monitoring*<br>• *Resource management* |
| Postconditions | *Tasks assigned and tracked* |

## UML-Activity Diagram



# 5. Diagrams

## 5.1 Use Case Diagram



| | |
|---|---|
| UC Name | Assign / Modify Passenger Seats |
| Summary | Airline staff shall assign, modify, or reallocate passenger seats for a flight. |
| Dependency | Flight schedule must exist; Passenger bookings must be present. |
| Actors | Airline Staff |
| Preconditions | Flight and passenger bookings exist. |
| Description of the Main Sequence: | • Step 1: Staff selects flight and views passenger list.<br>• Step 2: Staff assigns, modifies, or reallocates seats.<br>• Step 3: System updates seat assignments. |
| Description of the Alternative Sequence | • Step 1: Seat already occupied → system notifies staff and prompts for a different seat. |
| **Non-functional Requirements** | • System must process seat assignments within 2 seconds per passenger.<br>• Only authorized staff can modify seats. |

| | |
|---|---|
| UC Name | UC-31 Validate Passenger Ticket |
| Summary | Airline staff shall validate passenger tickets and confirm eligibility for boarding. |
| Dependency | Valid ticket database; Boarding eligibility rules. |
| Actors | Airline Staff |
| Preconditions | Passenger holds a valid ticket. |
| Description of the Main Sequence: | • Step 1: Staff scans or inputs ticket information.<br>• Step 2: System verifies ticket validity.<br>• Step 3: Staff approves or rejects boarding. |
| Description of the Alternative Sequence | Step 1: Ticket invalid or expired → boarding denied; system alerts staff. |
| **Non-functional Requirements** | • Verification must occur in under 1 second.<br>• System logs all validation attempts for audit purposes. |

| | |
|---|---|
| UC Name | UC-32 Manage Passenger Manifest |
| Summary | Airline staff shall create, update, and view passenger manifests for their airline's flights. |
| Dependency | Flight schedule and passenger booking data must exist. |
| Actors | Airline Staff |
| Preconditions | Flight exists. |
| Description of the Main Sequence: | • Step 1: Select flight.<br>• Step 2: Add, update, or remove passengers.<br>• Step 3: System saves and displays the manifest. |
| Description of the Alternative Sequence | Step 1: Missing or inconsistent data → system alerts staff and prevents saving. |
| **Non-functional Requirements** | • Manifest updates must be saved immediately.<br>• Only authorized staff can modify manifests. |

| | |
|---|---|
| UC Name | UC-33 Manage Boarding Process |
| Summary | Airline staff shall manage the boarding process, including scanning boarding passes and marking passengers as boarded. |
| Dependency | Valid tickets; Boarding procedures must be defined. |
| Actors | Airline Staff |
| Preconditions | Boarding started; passengers have valid tickets. |
| Description of the Main Sequence: | • Step 1: Scan boarding passes.<br>• Step 2: Verify passenger eligibility.<br>• Step 3: Mark passengers as boarded. |
| Description of the Alternative Sequence | Step 1: Invalid or duplicate pass → system alerts staff; passenger denied boarding. |
| **Non-functional Requirements** | • Boarding process must handle at least 100 passengers per 5 minutes.<br>• System logs all boarding events. |

| | |
|---|---|
| UC Name | UC-34 Generate Invoice |
| Summary | Finance staff shall generate invoices for landing fees, parking fees, and passenger services. |
| Dependency | Recorded airport services; Billing rules must be defined. |
| Actors | Finance Staff |
| Preconditions | Services recorded for billing. |
| Description of the Main Sequence: | • Step 1: Select services to invoice.<br>• Step 2: Calculate fees.<br>• Step 3: Generate invoice. |
| Description of the Alternative Sequence | Missing or incomplete service data → system alerts staff; invoice not generated. |
| **Non-functional Requirements** | • Invoice generation must complete in under 2 seconds.<br>• Only authorized finance staff can generate invoices. |

| | |
|---|---|
| UC Name | UC-35 Generate Financial Reports |
| Summary | Finance staff shall generate revenue, expense, and profit reports. |
| Dependency | Complete financial data; Reporting criteria must be available. |
| Actors | Finance Staff |
| Preconditions | Financial data available and up-to-date. |
| Description of the Main Sequence: | • Step 1: Select report type and period.<br>• Step 2: System compiles data and generates report.<br>• Step 3: Staff reviews and exports report. |
| Description of the Alternative Sequence | • Incomplete or inconsistent data → system alerts staff; report not generated. |
| **Non-functional Requirements** | • Reports must be generated in under 5 seconds.<br>• System ensures data integrity and confidentiality. |